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Web application user interface for customer relationship management in Spanish agritourism

BACHELOR'S THESIS

Study programme: Information media and services

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Abstract

This bachelor's thesis focuses on the design and implementation of the user interface and user experience for a web application aimed at customer relationship management in Spanish agritourism. The primary objective is to create a responsive and visually appealing front-end architecture that integrates with an existing backend system. This work forms a part of a larger project to develop a comprehensive digital solution for the agritourism sector.

The proposed application addresses challenges faced by small and medium farms with limited digital accessibility by providing tools for farm profile management, event creation, and visitor interaction. Insights for the development process are gathered through farm visits in Asturias, Spain, informal discussions with farmers, and an analysis of agricultural data to identify market trends and challenges.

The front-end is developed using modern web technologies such as Vue.js, Nuxt 3, and Quasar to deliver a high-performing and user-friendly interface tailored to the specific needs of agritourism. The application is validated by user testing, providing valuable insights for future improvements and validating the application's functionality and intuitiveness.

Keywords

Web application, agritourism, front-end, user interface, Vue.js, Nuxt 3, Quasar.

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Introduction

Agritourism combines agriculture and tourism and is an increasingly important sector in which farms can diversify their income, and visitors can experience rural life and sustainable practices. In Spain, a country of agricultural heritage and varied landscapes, agritourism can benefit both farmers and tourists. Nevertheless, the sector continues to experience challenges in reaching a broader audience because of an absence of digital platforms available to farmers despite its development. Many farms have a minimal or localized online presence, and potential visitors often need more information, limiting their ability to explore and take part in agritourism opportunities.

Farms open to visitors provide a range of different activities depending on what they have available or what is in season, ranging from berry picking to a contact zoo with farm animals such as sheep and cows. They provide educational encounters in animal care and environmentally conscious farming, which benefit urban families interested in locally produced, organic goods or tourists exploring the traditions of a particular area.

These paid visits also serve local farmers and communities. They add to farm income and allow farmers to diversify their revenue from traditional markets. Direct sales of produce create new revenue streams while supporting sustainable, local businesses. They also build community links through cultural exchange and appreciation of agricultural traditions among locals and visitors.

This sector is essential for rural development, allowing travelers to experience nature and agriculture traditions. Nevertheless, it is a potential goldmine - with problems including fragmented offerings, inconsistent digital accessibility, and few ways to connect farmers with potential visitors. Often, small family farms are also poorly visible and have limited promotional opportunities available to visitors. More awareness of those smaller farms will help the farm owners increase their reach and potential income, and the visitors will gain access to more varied and unique opportunities. This thesis's proposal of a web application is aimed at resolving such obstacles so that farm-based activities in Spain and potentially globally in the future can benefit rural communities and satisfy a growing need for nature-based experiences.

Goals of the Thesis

To achieve the main objective of the thesis and provide a relevant, useful, and user-friendly web application to the agrotourism participants, the work is divided into the following specific goals:

- Researching, analyzing, and validating agritourism trends and target users in Spain.
- Researching and analyzing existing solutions to define the value proposition of the new solution.
- Defining the problem and solution, identifying and analyzing the application system's user interface requirements.

- Designing and implementing user interfaces and user experience.
- Testing and refining the application system based on iterative internal testing and user testing.
- Evaluating and deploying the application.

Structure of the thesis

For a structured approach and documentation, the thesis is divided into the following chapters:

- Chapter 1 – Describes the importance of online presence for farms in the modern world by utilizing data from google trends and provides the list of relevant academic works that influenced and shaped this thesis.
- Chapter 2 – Describes the process of collecting and analyzing institutional agriculture data, key findings, and market evaluation.
- Chapter 3 – Describes the process of farm visits and discussions with farmers, analysis of existing solutions, and SWOT analysis with the intent to collect and define user requirements.
- Chapter 4 – Describes the proposition of the application utilizing use case diagrams, wireframes, and high-fidelity website prototypes.
- Chapter 5 – Describes the development process, application structure, utilized front-end frameworks, related web technologies, database structure, and how the front-end interacts with it.
- Chapter 6 – Describes user testing process and evaluates the results, providing insights for future improvements of the application.

Personal motivation

My journey into developing the front-end for agrotourism web application started when my family relocated to Asturias, Spain, on a business visa. During visits to Asturias, I was struck by the diversity and beauty of a Spanish landscape with much agricultural tradition. This initial fascination grew as we went to more farms, a lot of which provided us with guided tours and insights into their work. These experiences showed the potential of agritourism in the region, in which farms provide visitors with unique experiences while showcasing their produce and traditions.

Throughout these visits, though, a persistent challenge emerged. Information on available farms in the region was usually hard to get. Many farms did not have an online presence, a few had a limited localized reach, and some even had sites in languages that were not accessible to broader audiences. This digital gap gave rise to the idea of a platform designed to connect farmers with visitors and provide a structured, user-friendly interface to promote agritourism.

The concept gained further momentum when another family, relatives and friends of ours, joined the project. They contributed valuable expertise in website development, allowing the idea to take a more concrete form. With a fully assembled and motivated team, we formalized the concept into a comprehensive business plan. Recognizing its innovative potential, we submitted the necessary documentation and the plan to ENISA, a Spanish organization supporting entrepreneurship. The proposal was accepted, and we were granted a startup visa, officially enabling the development of application as a business venture.

This pivotal milestone coincided with an important decision in my academic journey: selecting a topic for my bachelor's thesis. I recognized this as an excellent opportunity to formalize my contributions to the project, integrating my academic objectives with practical experience. By documenting my role and the project's development process in this thesis, I aim to further the project's goals while demonstrating my commitment to its success.

1 Agritourism online presence and state of knowledge

This chapter aims to provide readers with a better understanding of how important it is for farms to have an online presence if they are willing to attract visitors. Since visitors and tourists most commonly utilize the web search to find entertainment in the area, this is supported by data obtained from Google Trends. The second part describes which relevant academic works that influenced and shaped this thesis were researched and studied.

1.1 The importance of online presence in agritourism

Nowadays, people are increasingly moving into the online environment. Establishing a website for farm visits is not just a trend but is becoming a necessity for farm owners for several reasons. The first reason is to reach the target audience. Most visitors today search for information about farm experiences and activities online, as shown in the graphs Figure 1 and Figure 2. The data was obtained using the Google Trends website, inputting the desired search term and setting further filters. The numbers on the Y axis indicate search interest relative to the highest point on the chart for the specified region and time. A value of 100 represents peak popularity for the term, while a value of 50 shows the term is half as popular. A score of 0 signifies insufficient data for this term.

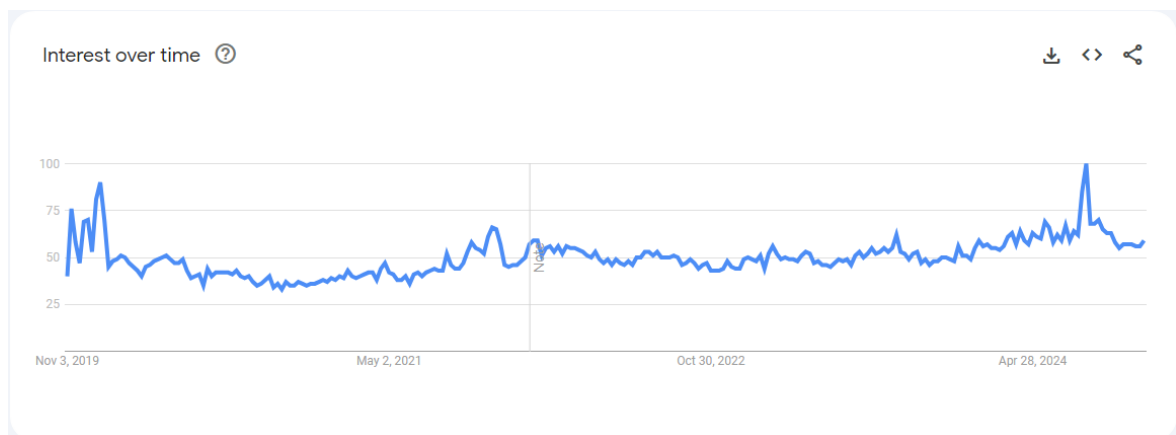


Figure 1: Popularity of the search term "Farm" (source: Google Trends, 2024)

The graph represents the data obtained using the following filters: Region - Spain, Time range - Past 5 years, All categories, Web search.

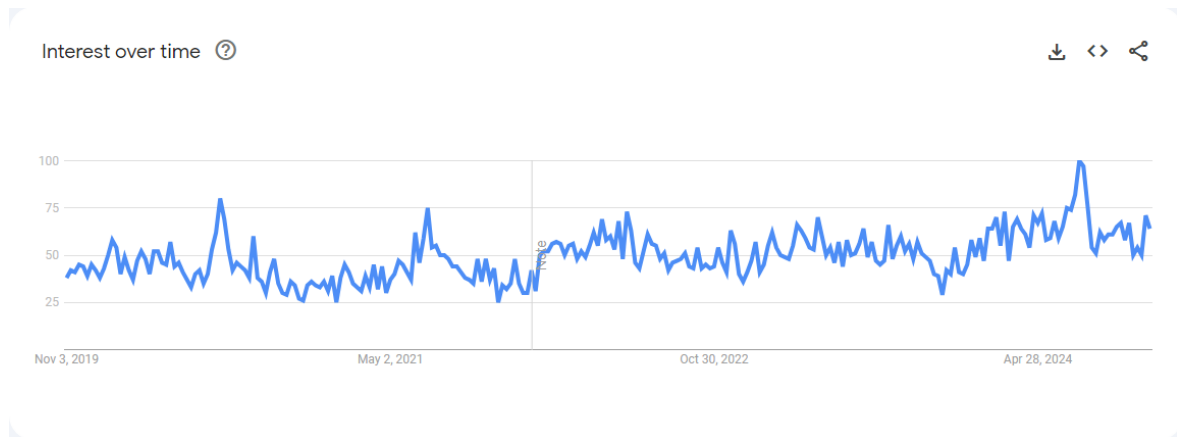


Figure 2: Popularity of the search term "farm Spain" (source: Google Trends, 2024)

The graph represents the data obtained using the following filters: region - worldwide, time range - Past 5 years, all categories, and web search.

A website provides farm owners with a way to present information about activities, seasonal events, prices, schedules, and conditions for visiting the farm. Such an approach helps attract visitors and persuade them that their farm is the desired experience.

Further considerations are convenience and efficiency for visitors and farm operators. Having a website eliminates physical visits, phone calls, or mailed requests to obtain information about the farm offerings. The central platform reduces the administrative burden for farm owners by combining visitor information and communications.

The web platform also allows easier interaction with visitors during their farm experience. Online updates may include activities, photos, or health and safety information.

Furthermore, an online presence can also be marketed. A website allows farm owners to promote their activities more effectively across various digital channels, such as search engines or local event listings, to attract interest in new seasonal offerings and updates available on the website.

1.2 Collection of relevant academic works

The following academic papers were identified using research tools such as Google Scholar, theses.cz, researchgate.net, and other academic databases. Keywords such as "agritourism," "web application," "frontend framework," "Vue.js," and "responsive design" were utilized to locate relevant studies. The selected works were researched for supporting information to achieve the objectives of this thesis, which will serve as a source of inspiration and knowledge for the development of the proposed web application.

Lyxel, Oskar. Server-Side Rendering in React: When Does It Become Beneficial to Your Web Program?

This dissertation is an experimental study aimed at presenting knowledge about the advantages and disadvantages of rendering methods. In this research, two identical web applications containing a grid were created. It provides readers with an overview where SSR, compared to CSR, generates a positive value in the form of increased performance over time, which was measured through a sequence recording combined with user experience. (Lyxel Oskar, 2023)

Derleth, Thomas. Corporate Component and Service Libraries - A concept for creating, maintaining and managing a company-specific user interface component library in the field of front-end web development.

The main purpose of this thesis is to provide a general view of the current state of the art in research on important front-end framework characteristics. In particular, the author provides the reader with specific qualities that are important for evaluating a front-end framework for a software project. The author explains the characteristics of these frameworks and the reasons for their creation. Furthermore, he demonstrates a component-oriented architecture for user interface development and evaluates it in terms of three quality criteria - user interface consistency, performance, and modifiability. An interesting element of the work is the development of a structured method for evaluating a framework and providing examples of its use, where the author tries to guide the reader on how to choose the right framework for a project. (Thomas Derleth, 2018)

Rude, Simon. Implementation of the data layer of the front-end application Adgame and communication with the back-end.

This thesis is dedicated to the implementation of the data layer and communication with the back-end in the front-end of the Adgame application. It provides readers with an overview of the tools, methods, and libraries for web application development. In the practical part, the author examines the development of a web application, creates an interface for interaction with a REST API, evaluates the performance of modern tools and application frameworks in practice, and describes the importance of a centralized data layer in the form of a Vuex store, justifying and illustrating its division into various modules. After a basic description of data handling, the work explains the overall process of integrating the obtained data into existing Vue.js components and illustrates the challenges that arise during this process. Readings from this thesis will be particularly beneficial to this bachelor thesis during the implementation phases. (Rude Simon, 2021)

P. Šimek, J. Jarolímek, J. Masner. Cross-Platform User Interface of a Web Application in Agrarian Sector.

The article treats the process of the creation of a web application optimal output for mobile devices in the form of a responsive layout with focus on the agrarian web portal. The utilization and testing of user experience techniques in four steps - UX, research, design and testing - were of great benefit. The resulting responsive layout was developed with the emphasis on the ergonomic layout of control elements and content, a conservative design, the securing of content accessibility for disabled users and the possibility of fast and simple updating. The resulting knowledge is applicable to web information sources in the agrarian sector (agriculture, food industry, forestry, water supply and distribution) and the development of rural areas. In wider context, this knowledge is valid in general. (P. Šimek, J. Jarolímek, J. Masner, 2015)

Jasiński, Adam. Report from the study: Effectiveness of promotion using web-based agritourism portals in the light of their users' feedback. Web portal model for the region.

This article deals with the evaluation of the effectiveness of promotion through agritourism internet portals, considering user opinions. To provide a more comprehensive view of the research, the portals were analyzed from three perspectives: content analysis, opinions of agritourism farm owners, and opinions of individuals interested in utilizing the offerings of agritourism farms and obtaining information about them through internet agritourism portals. The author also presents modern recommendations for expanding the use of new information technologies and for promoting agritourism farms to develop comprehensive agritourism. (Adam Jasiński, 2012)

2 Problem definition and market availability analysis

Understanding agritourism from afar requires first understanding trends in farming and relevant data. National agricultural data, such as census reports, give insight into how farming communities, land use, crop types, and economic conditions shape agritourism in indirect ways. By analyzing data on farm sizes, crop diversification, organic practices, and demographic changes in farm ownership, this analysis identifies the underlying changes in agriculture supporting the agritourism sector.

Such data points highlight agritourism's growth as well as the challenges and opportunities for small-scale farms that are often involved in agritourism. Further chapters justify how these agricultural trends correspond to and validate the growing demand for farm-based visitor experiences.

2.1 Sources

Agriculture trends were validated by analyzing reliable data sources. Key among these is the Agricultural Census which provides information on farm structures, land use, crop types, and changes in agricultural practices in Spain. The latest census, which was carried out in 2020 and published in May 2022, identifies trends that indirectly affect agritourism: small farms, organic farming, and regional variations in crop production. Additionally, Eurostat's official website, which provides broader European agricultural statistics, was consulted. Further, the Farm Accountancy Data Network (FADN) was used to analyze the economic conditions of agricultural holdings in the EU. These sources give an overview of the agrarian landscape and place agritourism trends within broader trends in farming and rural economies.

2.2 Key Findings

The following subchapters present key findings derived from the analyzed data, followed by hypotheses formulated based on the identified trends.

2.2.1 Decreasing number of small farms

Table 1: Changes in Agricultural Holdings and Utilized Agricultural Area (2009–2020) (source: INE)

	2009	2020	% Variation
Number of agricultural holdings	989,796	914,871	-7.6
Utilized Agricultural Area (UAA) Unit: Hectares	23,752,688	23,913,682	0.7
Average UAA per holding	24.56	26.37	7.4

The data indicates a 7.6% decrease in the number of farms, with 70,000 fewer farms compared to the previous edition in 2009. Meanwhile, the utilized agricultural area has increased by 0.7%, surpassing 23.9 million hectares. This results in a higher average farm size, reaching 26.37 hectares—7.4% larger than in 2009.

The trend toward farm specialization persists, as reflected in the rise in average utilized agricultural area (UAA) per farm and a decline in the total number of agricultural holdings, in addition to the growing prevalence of management by legal entities rather than individual owners.

As larger industrial farms become more prevalent, smaller farms may find it challenging to distinguish themselves in an increasingly competitive environment. A website dedicated to featuring small, welcoming farms would offer essential exposure and assistance.

2.2.2 Increased area dedicated to organic farming

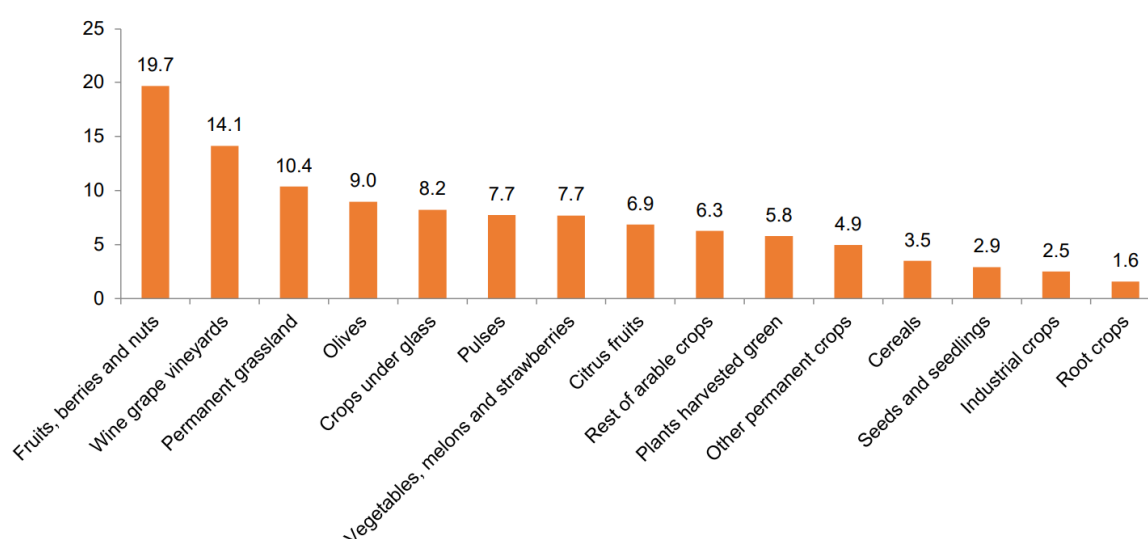


Figure 3: Percentage of organic UAA of total UAA by crop type (source: INE)

In 2020, organic utilized agricultural area (UAA) represented 7.8% of total UAA in response to a trend toward sustainable and eco-friendly farming. Some crop types had a particularly high percentage of organic farming in terms of total cultivated area. The highest percentage of organic UAA was fruit trees and berries (19.7%) and wine grape vineyards (14.1%). In contrast, crop types such as roots and tubers (1.6%) and industrial crops (2.5%) accounted for the lowest share of organic UAA, suggesting less organic use in these categories.

This data illustrates a clear shift towards organic cultivation, especially for high-demand fruit and vineyard crops, which are the best crops for farms that are welcoming visitors and tourists and in line with consumer demand for sustainable produce. The expansion of organic farming means that smaller, specialist farms can take advantage of this demand.

2.2.3 Prominence of Fruits, Berries, Nuts, and Olives in Organic Farming

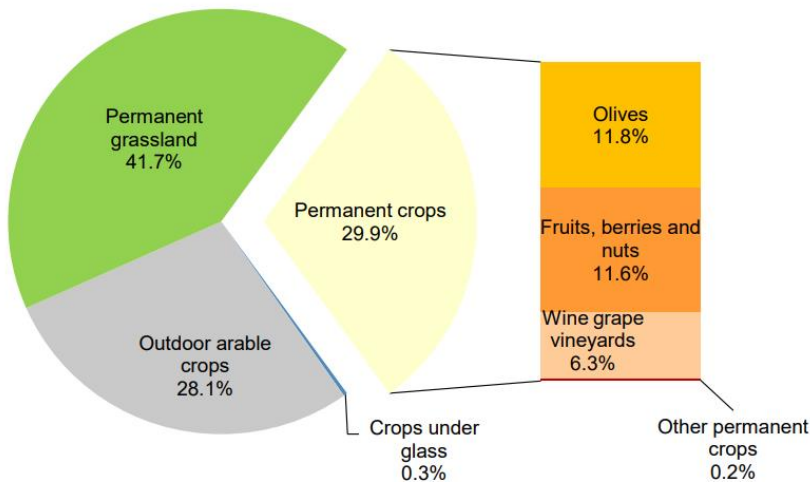


Figure 4: Percentage of organic UAA of total UAA by crop type (source: INE)

Within the organic farming sector, some crops stand out due to their high share of the total organic utilized agricultural area (UAA). Fruits, berries, olives, and nuts together constitute a substantial share of organic crop production, reflecting their importance for consumer demand and sustainable agriculture. Olives account for 11.8% of the organic UAA, followed by fruit trees and berries at 11.6%. This concentration highlights the staple nature of these crops in organic farming, driven by domestic as well as international demand for organic and sustainably produced goods. Overall, the utilization of agricultural area (UAA) for fruits, berries, and nuts increased by 23.8% from 2009 to 2020, but this growth was not only organic. In the overall agricultural sector, the UAA for fruits, berries, and nuts increased by 23.8% between 2009 and 2020.

The high proportion of these crops in organic farming also corresponds to the growing interest in agritourism. Farms with produce (fruits, berries, nuts) are ideal for "pick-your-own" experiences that appeal to visitors looking for hands-on, seasonal interactions with nature.

2.2.4 Agricultural work on the farm

Table 2: Farm labor force, Unit: Total Annual Work Unit (AWU) (source: INE)

	2009	2020	% Variation
Total	922,026	851,405	-7.7
Holder	330,909	318,520	-3.7
Family labor force	232,774	116,828	-49.8
Contracted labor force	325,286	378,392	16.3
Subcontracted labor force	33,057	37,665	13.9

The figures reveal a 7.7% reduction in the total agricultural workforce in 2020 from 2009. There were 3.7% fewer farm owners working on their farms, and the family labor force declined by nearly half (49.8%). This contrasted with rises in contract workers (16.3%) and subcontracted labor (13.9%).

This substantial reduction in family labor may indicate difficulties in maintaining family-owned farms that traditionally rely on relatives to support operations. This trend may signal the need for family farms to find alternative income streams, and a platform with greater visibility and visitor engagement could provide revenue support.

On the other hand, the increase in contracted and subcontracted labor indicates the increase in seasonal and permanent labor, potentially for larger farms. This need could be met by the planned job listing and search features within the proposed application platform that would connect available labor to farm employers and meet workforce demands across diverse farm types.

2.2.5 Gender Diversity in Farm Management



Figure 5: Percentage variation in the number of managers by gender and type compared to 2009 (source: INE)

Total farm manager employment dropped by 7.6% in 2020 compared with 2009. Despite this overall decrease, female farm manager representation increased. Women farm managers increased by 22.0% between 2009 and 2020 to 28.6% of all managers. In contrast, the number of male managers declined by 15.8%. In addition, when the farm owner differs from the day-to-day manager, female managers increased 72.2% compared to male managers (26.4%).

This gender shift reflects a changing demographic in farm management, where women are increasingly leading farm operations. This increased diversity presents an opportunity to promote inclusivity in agriculture. A site devoted to farm visits might feature women-led farms, demonstrating diversity in farm management and appealing to visitors who support inclusive and community-driven agriculture.

2.3 Market valuation and expected evolution

A hypothesis about the market size is necessary to assess the early market potential of a business. Startups and emerging enterprises often use the PAM, TAM, SAM, and SOM frameworks to estimate market size and growth prospects. This determines whether a project is worthwhile financially. These calculations are often shown to potential investors in initial discussions about funding. Here, data for estimating the market will come from the Farm Accounting Data Network (FADN). Estimated values will be focused only on hosts (farmers) since reliable data on this group is available, rather than more variable and harder-to-obtain information about potential users (visitors).

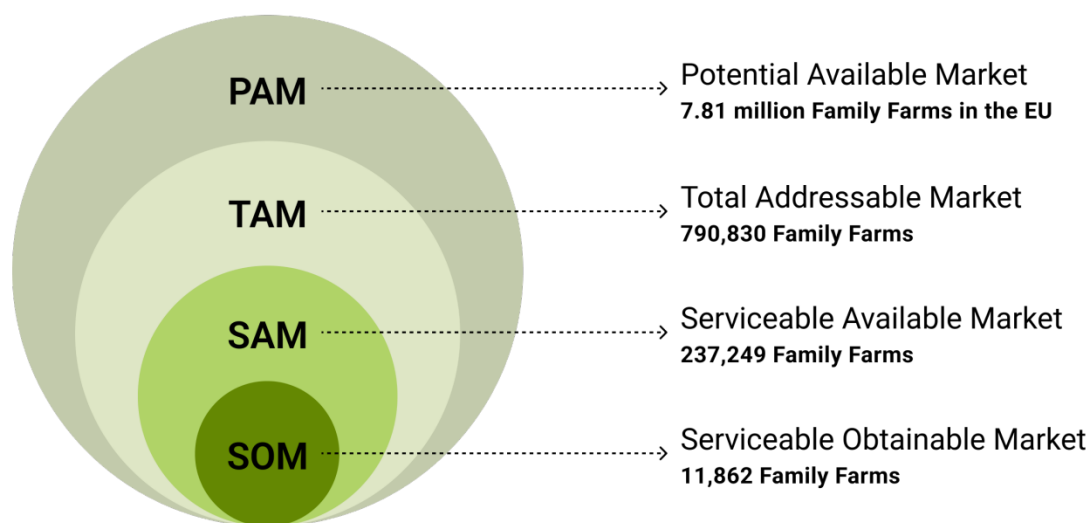


Figure 6: Market availability (FADN data, Author's calculation)

- **PAM (Potential available market)**

Considering expansion across other European countries, focusing on family farms. Estimation: Out of the 8.98 million farms in the EU, approximately 7.81 million are family farms.

- **TAM (Total addressable market)**

This category represents the total family farms in Spain. Family farm numbers are used as references since those would benefit the most from a dedicated web platform and are less likely to have personal websites already.

Calculation: In Spain, 87% of the 909,000 farms are family farms, resulting in approximately 790,830 family farms.

- **SAM (Serviceable available market)**

Considering not all family farms will have the same level of online accessibility or interest in an online platform.

Estimation: Targeting 30% of the TAM would amount to 237,249 family farms.

- **SOM (Serviceable obtainable market)**

The practical segment of the SAM realistically aimed to be captured in the initial phase.

Estimation: Assuming 5% of the SAM are captured, this amounts to 11,862 family farms. Targeting family farms with average and below-average income, the proposed application has a significant market opportunity both within Spain and in the broader European context.

3 Analysis of user demands and definition of proposition value

User requirements are an essential step of system development to make sure that the end result fulfills the demands and expectations of the intended users. Identifying and prioritizing these requirements enables developers to develop a functional and easy-to-use system that could solve particular user issues. This approach reduces the risk of misaligned features and increases user satisfaction and adoption.

This chapter is dedicated to the identification of the needs of farmers, the key users, and the primary sources of content for the agritourism platform. Farmers contribute profiles, events, and offerings that add value to the platform for visitors, making their needs central to the system's design.

Without scientific data or formal user studies, the analysis presented in this chapter is based on the development team's experience and observations during visits to different farms in Asturias, Spain. These visits also included informal discussions with farm owners about their operations, challenges, and potential digital needs. Not exhaustive nor statistically rigorous, this approach provides a practical understanding of user requirements based on real-world interactions.

3.1 Method of User Requirement Analysis

Gathering user requirements for this platform required practical, hands-on methods to gain a deep understanding of the needs and challenges faced by farmers in agritourism. Together with the proposed application team, the author has participated in visits to over ten farms in Asturias, Spain, where we explored their operations, engaged with their activities, and spoke directly with the farmers. These visits allowed to participate in farm activities, see produce, and also have informal discussions with farm proprietors. The length and depth of these conversations depended on if the farmers were willing and available to speak.

Three farms have been chosen for thorough analysis because of the richness of information they provided: “El Malain” (a family-oriented farm with a focus on direct sales and fruit picking), “Finca Ecológica Terramora” (a farm specializing in growing mini kiwis and instructing visitors on sustainable farming methods), and “Los Caserinos” (a family-owned farm specializing in producing high-quality dairy products and providing interactions with animals).

These farms have been selected as examples to give various points of view on the needs of agritourism platforms.

This approach was chosen over more formal scientific techniques under several practical constraints. Time limitations for structured studies, including surveys or focus groups, across a larger sample of farmers. Additionally, farm owners' availability and their willingness to participate in detailed research posed challenges, making informal, opportunistic discussions during farm visits a more feasible option.

Alternate user research techniques such as structured questionnaires, focus groups, and observational studies were considered but have yet to be implemented. Such a technique would have required more detailed planning, resources, and formalized cooperation from more farmers than could have been accommodated in this project. However, insights gained through direct interaction with farmers during visits provided a practical and contextually rich understanding of their needs that served as the basis for defining system requirements.

3.2 Profiles of Selected Farms

To gain a deeper understanding of the diverse needs of farmers in agritourism, three specific farms from Asturias, Spain, were selected for detailed analysis. These farms were selected by their specific features, their approaches to agritourism, and their insight into the challenges of farming operations of different types.

3.2.1 El Malain

El Malain is a family farm focused on agritourism. The farm lets visitors pick fruits such as berries and apples directly from the orchards. El Malain also runs a small farm shop where visitors can buy fresh produce such as ice cream jams and other local products.



Image 1: El Malain farm (source: author)

Key Activities:

- Fruit picking experiences for visitors.
- Selling fresh and processed products (e.g., jams, cider, ice-cream) at the farm shop.
- Hosting seasonal events, such as harvest festivals.

Challenges:

- Managing and streamlining events for fruit-picking activities, especially during peak seasons.
- Effectively promoting seasonal events to attract visitors.
- Ensuring consistent visitor engagement and satisfaction.

3.2.2 Finca Ecológica Terramora

Finca Ecológica Terramora is an organic farm where mini kiwis are grown. The farm is sustainable and also provides visitors with educational opportunities to find out about organic farming methods and the various aspects of their produce. Its operations are designed to be eco-friendly and community oriented.



Image 2: Finca Ecológica Terramora farm (source: author)

Key Activities:

- Cultivating and selling organic mini kiwis.
- Hosting lectures and workshops about this fascinating fruit and the business surrounding its cultivation.
- Providing volunteer opportunities for individuals interested in organic agriculture.

Challenges:

- Expanding the reach of their workshops and lectures to attract a broader audience, particularly through online promotion.
- Creating engaging digital content that highlights the unique aspects of their farming practices and products.

3.2.3 Los Caserinos

Los Caserinos is a family-run farm that specializes in producing high-quality dairy products, including cheeses and yogurts. The farm also offers interactive visitor experiences, such as guided tours and opportunities to interact with farm animals.



Image 3: Los Caserinos farm (source: author)

Key Activities:

- Producing and selling artisanal dairy products such as cheeses, yogurts, and milk.
- Organizing guided tours of the dairy facilities to demonstrate the production process.
- Allowing visitors to interact with farm animals as part of the experience.

Challenges:

- Enhancing online visibility and digital marketing to reach a wider audience.
- Streamlining visitor bookings and tour management to accommodate both individuals and groups.

3.3 Review of existing solutions

This section focuses on analysing the existing solutions within the Spanish agritourism market. Examining such solutions aims at identifying gaps and opportunities for enhancement, which our proposed web application could fill. Detailed competitive analysis allows us to craft an optimal value proposition to meet and even exceed the demands of today's agritourists and farm hosts. This strategically placed approach makes the proposed solution stand out in the market, offering features that are new and directly related to user demands.

3.3.1 Farm Stay Planet

Farm Stay Planet is a global platform dedicated to connecting users with agritourism accommodations, specializing in farm stays. The site serves primarily as a search tool for its users, offering various functionalities to assist travelers in finding farm accommodations worldwide. While it offers search services, the actual booking process is facilitated through the external platform, Booking.com, reflecting a business model based on affiliate partnerships. According to the site's publicly stated information, its revenue stream derives solely from commissions earned through these bookings.

Farm Stay Planet has committed to supporting charitable organizations, notably The One Acre Fund, which assists smallholder farmers in East Africa, and Farm Forward, which advocates for the reduction of animal suffering in the agricultural sector. In 2020, the site also initiated a partnership with onetreeplanted.org, where one tree is planted for every booking made through their platform on Booking.com. (Farm Stay Planet, 2024)

The website offers search via an interactive map, keyword search, or location-based search with input from nearby towns or cities, but not country-specific searches. Farm Stay Planet lacks a user profile creation feature - a reiteration of its role as a search engine rather than a fully integrated customer management platform. The listing page for each farm includes basic descriptive details like the property description, photos, and a visual map representation. Users can also see farm availability and pricing but are directed to Booking.com for

the actual booking. That structure illustrates how the site depends on outside booking platforms to function.

The site's navigation is simple: by country, continent, and category. Every country page has local agritourism information and a list of recommended farm stays. However, it supports only two languages -English and Italian which may make the site unsuitable for non-English or Italian users.

In terms of user experience, the design and layout follow a functional approach for users looking for simplicity when searching for farm accommodation. However, redundancy between the navigation bar and search options may signal areas for possible optimization of the user interface, as this duplication does not significantly improve the experience. Despite those limitations, Farm Stay Planet's role as a directory of Farm Stay accommodations and its philanthropic efforts make it a noteworthy player in agritourism.

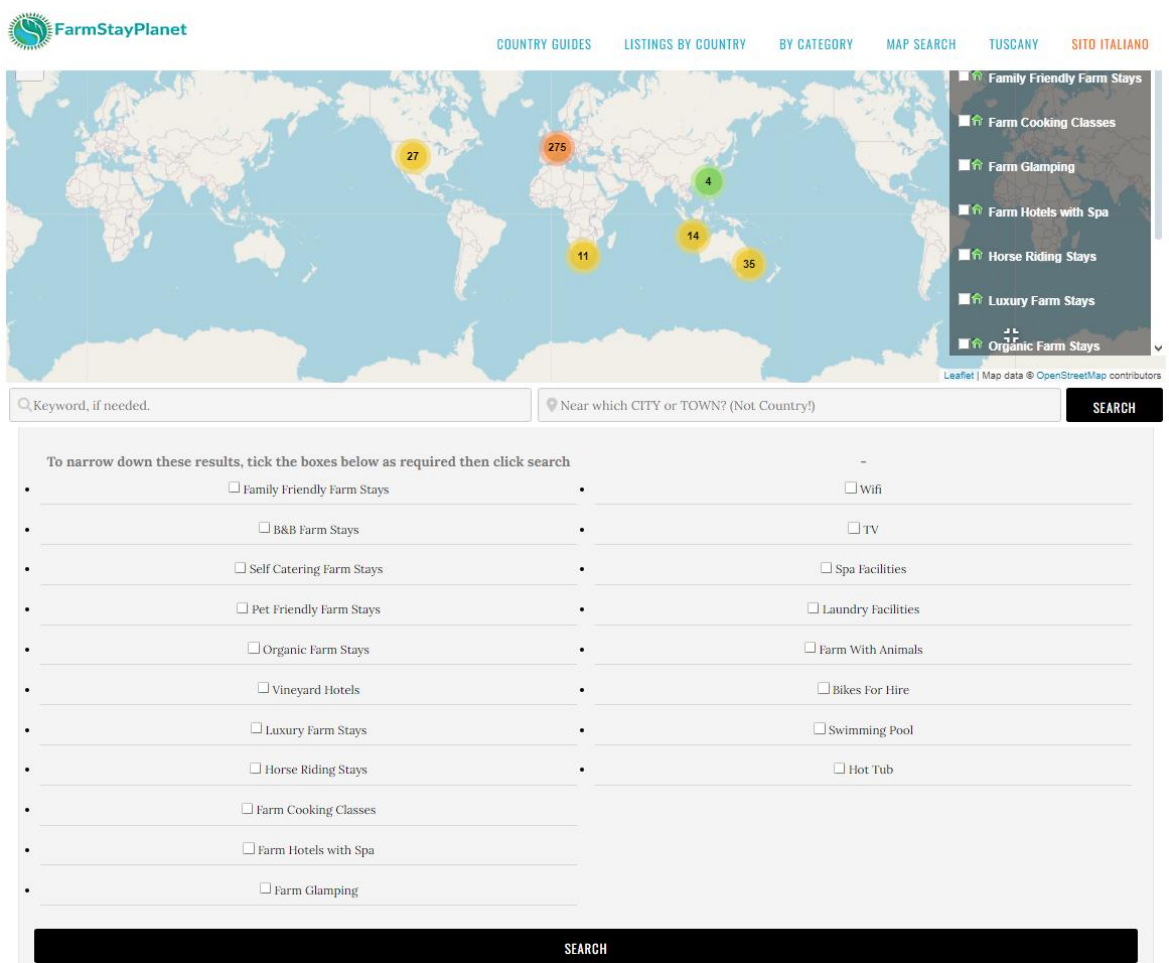


Image 4: Farm Stay Planet Web (source: Farm Stay Planet retrieved October 29, 2024)

3.3.2 Ruralidays

Ruralidays is an online platform specializing in rural vacation rentals, mainly focused on southern Spain, with a wide selection of properties across Andalusia. The website offers a seamless booking experience, allowing users to search, compare, and book vacation homes

directly through the platform without relying on third-party services. Its functionality includes advanced filtering options, allowing users to refine their searches based on location, amenities, number of guests, and specific features such as pools or pet-friendly properties. Each property listing is detailed, providing descriptions, high-quality images, customer reviews, and real-time availability.

The user experience and interface of Ruralidays are intuitive and user-friendly. The website features a clean and modern layout with easy navigation, making it simple for users to browse and filter properties. The search functionality is straightforward, and the booking process is smooth and fast, benefiting from clear calls to action. Additionally, the platform supports multiple languages, including English, Spanish, French, and German, catering to an international audience and enhancing its accessibility.

Some of the platform's advantages include extensive property listings in rural Spain, detailed property descriptions, and real-time availability. The company's customer support services before, during, and after the booking process give users additional peace of mind. The website is also multilingual supported, making it more competitive in the rural tourism market.

However, some disadvantages include its geographical limitation to Spain and Andalusia, which might restrict its appeal to those looking for rural stays outside of the country. Additionally, while the website's layout is generally intuitive, users seeking more unique or specific features beyond standard filters might find the search options somewhat limited. Those are minor drawbacks, but Ruralidays remains a solid player in the niche of rural vacation rentals, offering a simple-to-use platform that puts convenience and trustworthiness first for its users.

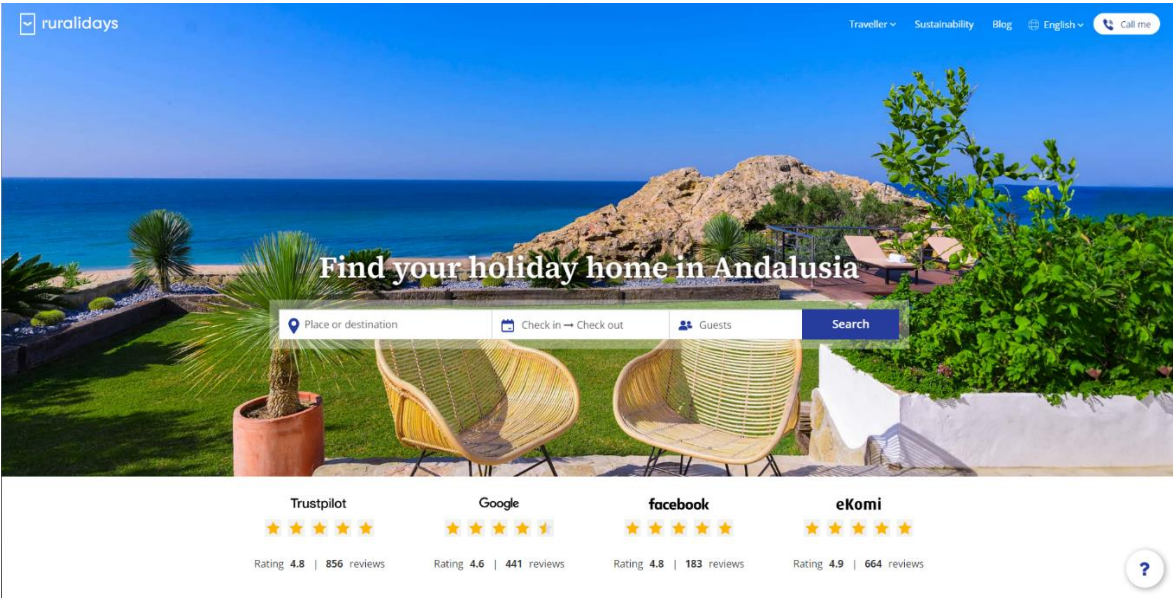


Image 5: Ruralidays Web (source: Ruralidays, retrieved October 29, 2024)

3.3.3 WWOOF

WWOOF (Worldwide Opportunities on Organic Farms) links volunteers with organic farms in Spain to promote sustainable agriculture and cultural exchange. Voluntary workers are called "WWOOFers" and can work on planting, harvesting, and eco-friendly construction in return for accommodation and meals. For those kinds of opportunities, users must subscribe to the platform and pay USD 20 for individual membership or USD 36 for dual membership annually. (WWOOF, 2024)

Farm hosts offering volunteer opportunities also pay USD 30 annually to list their farms and connect with volunteers. Though WWOOF Spain is a non-profit organization that claims that "no money is exchanged between hosts and WWOOFers," the funds collected from volunteers' and hosts' subscriptions are not clear as to how the fees are distributed or used. There is little information available about the distribution or purpose of those fees. (WWOOF, 2024)

In April 2024, WWOOF redesigned its website to replace its earlier, less appealing version. Until these updates, WWOOF Spain was a smaller competitor in the agritourism and organic farming sector. Yet these recent improvements have also included many features that were initially planned for proposed application - like better search tools, more straightforward navigation, and just an overall cleaner look and feel.

The updated WWOOF Spain platform now includes advanced search filters for browsing farms by location, type of activities, and specific tasks. Each farm listing describes the work,

the cultural experience, and the accommodation type. The site also facilitates direct communication between volunteers and farm hosts, making it easier to coordinate stays and tasks.

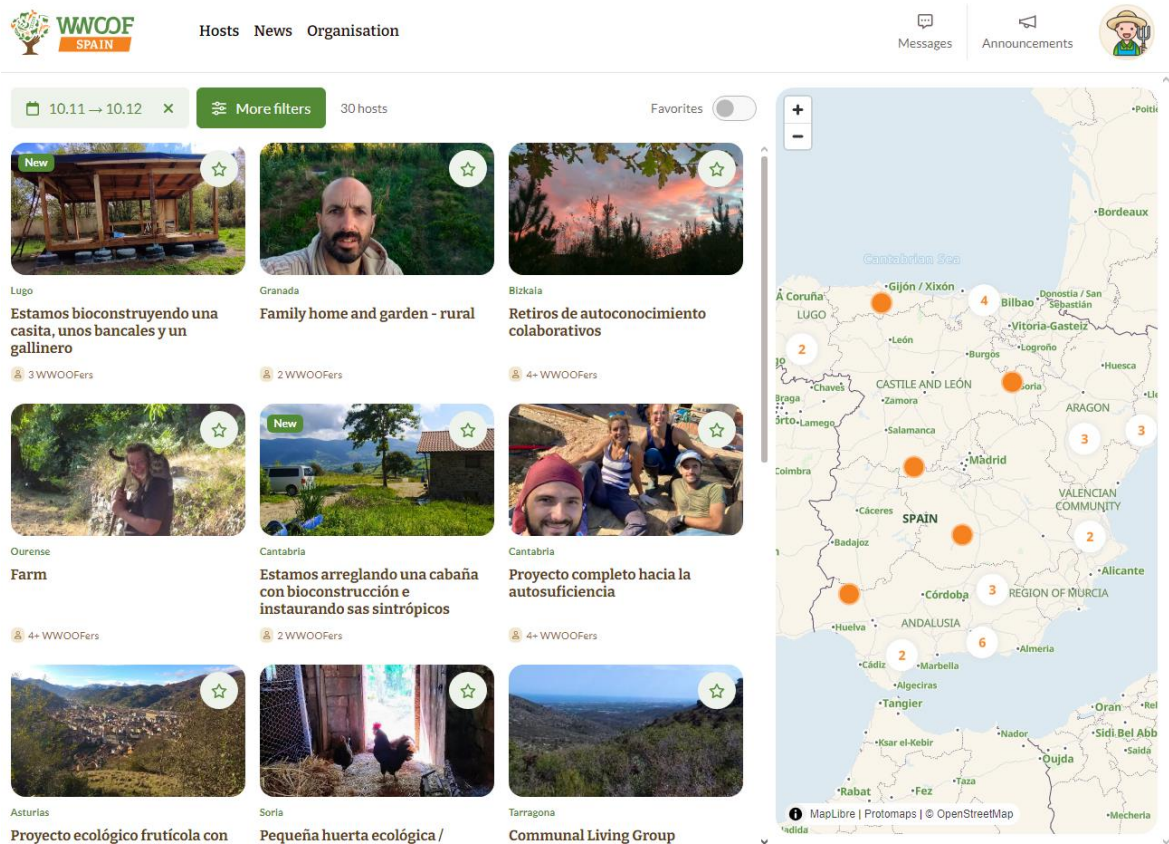


Image 6: WWOOF Hosts Page (source: WWOOF, retrieved October 28, 2024)

3.3.4 Summary, comparative evaluation

After a thorough evaluation of the existing solutions in the agritourism market, several key areas for improvement have been identified, alongside essential features that should be incorporated into the proposed web application. Current platforms serving similar purposes offer valuable insights into the strengths and weaknesses prevalent in the industry. These insights provide a clear direction for enhancing user experience, functionality, and market positioning in the proposed solution.

The following table compares the features, values, target audience, and revenue model of the examined solutions and of the proposed solution.

Table 3: Overview of existing solutions (source: author)

	Farm Stay Planet	Ruralidays	WWOOF Spain	Proposed application
Target Audience (Guests)	Travelers, nature lovers	Tourists seeking rural holiday rentals	Volunteers looking for farm work in exchange for a stay	Travelers, Agritourists, Job and knowledge seekers
Target Audience (Hosts)	Farms focused on eco-friendly organic production	Accommodation providers	Farms providing education and accommodation	Agritourism businesses, small family-owned farms, seasonal job providers
Geographic Focus	Global	Spain, Andalusia	Global WWOOF network	Spain, later global
Booking Process	External Through Booking.com	Integrated Booking	Integrated communication with hosts	Integrated communication tools
Review and rating system	no	Only hosts reviews and rating	no	Both hosts and guest reviews and rating
Search Functionality	Interactive map, keyword, filters	Advanced search filters	Location, dates, and number of guests	Location, interactive map
Revenue Model	Commission on bookings	Commission on bookings	Annual subscription 20\$ guests, 30\$ hosts	Credit-based internal economics
Multilingual Support	English, Italian	Yes	Yes	Yes
Unique Value Proposition	Specialized in promoting farm stays globally	Focus on holiday rentals	Cultural and educational exchange through farming	Differentiating features and user-centric approach

3.4 SWOT Analysis

In this part, SWOT analysis is utilized to evaluate the current state of the agritourism sector and the proposed platform. SWOT analysis is a method of developing conditions and evaluating a problem, project, or business concept based on internal and external factors, namely strengths, weaknesses, opportunities, and threats. This method is most often used in business evaluation methods to find strategies that will be carried out. SWOT analysis only describes the situation that occurs, not providing the solution to the problem. SWOT analysis is carried out with the intention of identifying the level of readiness of overall functions carried out to achieve the goals that have been set.

3.4.1 Strengths of the proposed system:

- **Targeted features:** The platform offers job listings, product listings, automatic farm description generation, and event creations, which can be useful for potential applicants in job search and farmers for selling their products and promoting their events.
- **Customization of profile and farm page:** Farmers can customize their profiles and farm pages to showcase their products and services in a way that appeals to their target audience and differentiates them from competitors.
- **Automated email delivery:** The platform offers an automated email send feature, which can help farmers save time and effort by sending promotional emails to potential customers and encouraging them to use the farm's services.
- **Enhanced visibility for small farms:** The platform offers smaller farms a platform to gain exposure and compete with larger industrial farms.
- **Accessibility to users and farm owners with less technical skills:** The platform is designed to be user-friendly and accessible to farmers with less technical skills, making it easy for them to navigate and use the platform.

3.4.2 Weaknesses of the proposed system:

- **No booking directly through the website:** Unlike other popular platforms such as Airbnb, Booking.com, and WWOOF, which allow users to book directly through their websites, proposed web application does not offer this feature, however it is an intended system design, since some farms may have established booking system while others may not require it at all.

- **Limited Initial Reach:** Requires marketing efforts to attract farm owners and users to the platform during the initial launch phase.
- **Dependency on Internet Accessibility:** Relies on farm owners and rural users having stable internet access, which may be limited in some areas and can make web applications vulnerable to technical issues and outages, which can negatively impact customer experience.

3.4.3 Opportunities of the proposed system:

- **Growth of Agritourism:** Increasing interest in sustainable and experiential tourism provides opportunities to expand the platform's features to attract a wider audience.
- **Expansion of Features:** Adding multilingual support, integrated payments, or job boards could make the platform more accessible and convenient for users.
- **Rising Organic and Local Trends:** Highlighting organic and small-scale farm offerings could help the proposed application to appeal to environmentally conscious consumers, attract a new audience, and differentiate itself from other travel platforms.

3.4.4 Threats of the proposed system:

- **Competition from other platforms:** The platform may face competition from larger platforms like Airbnb or niche agritourism websites that cater to farmers and dominate the market with their established user base and resources, which can limit its market share and impact its ability to attract and retain customers.
- **Seasonal Demand:** Agritourism activities are often seasonal, potentially leading to inconsistent traffic on the platform, which can impact the platform's ability to attract and retain customers.
- **Technological Barriers to Target Audience:** Rural farm owners may be hesitant to adopt new technology due to their lack of familiarity with platform work and the potential risks associated with it, which could limit the effectiveness of the web application and limit from reaching its full potential.

3.5 Requirements

The following requirements were identified for the proposed application; they will be divided into three parts - general requirements (for all users), farmer-specific requirements, and visitor-specific requirements.

General requirements:

- Profile creation, signup, and login
- Profile maintenance (changing photos, personal information, description)
- Multilingual support
- Feedback and review system
- Communication tools
- Responsive design
- User-friendly interface
- Data security

Farmer specific requirements:

- Farm profile and page creation
- Farm page maintenance (changing photos, descriptions, contact details, activities, offerings, addresses, events, and job listings)
- Multilingual description generation tools
- Event publishing and promoting tools
- Displaying available products
- Providing contact information
- Browsing visitor reviews

Visitor specific requirements:

- Farm search functionality
- Farm browsing functionality
- Saving and removing farms and events to the favorites list
- Visual representation of farms and events location
- Visual representation of farm owner's contact details
- Contacting the farmer
- Selecting language preferences
- Providing and accessing reviews and ratings

4 Proposal and System design

This chapter provides the list of features that are planned for the application based on the analysis in prior chapters 2 and 3. Further, it describes the Use-case diagrams, wireframes, and high-fidelity prototypes of various proposed application pages.

4.1 Differentiating features

To effectively address the challenges identified in the agritourism sector and create a competitive advantage, the proposed web application is planned to include a set of differentiating features aimed at delivering value to customers, the market, and the broader economy. The functionalities of future web application are derived from information gathered from the analysis of agricultural trends and user requirements, as described in prior chapters 2 and 3. These studies discovered gaps in the digital accessibility and connectivity of agritourism services along with the various needs of farmers and visitors. By leveraging these findings, planned functionalities include improving the online presence of farms, facilitating communication between stakeholders, and improving user experience. The features are intended to satisfy the particular requirements of medium-sized and small farms in order to compete with more significant enterprises and allow visitors to explore and participate in agritourism opportunities. This chapter discusses the key functionalities that are planned to be implemented in the platform, which are in line with identified user needs and have the potential to drive growth and sustainability in the agritourism sector.

Events and community calendar

The platform is planned to include a feature allowing farmers to list special events, such as weekend markets and events or offerings specialized to their farm. This information will be accessible to all visitors so that users can learn about and attend these events, fostering greater engagement and participation in the agritourism community.

Profile creation for farmers

The proposed application is aimed to provide an easy-to-use interface for farmers to create a web representation of their farm. This will allow farmers to describe their farm in various languages and present information about their produce or livestock, activities available, accommodation facilities, and any special events or classes they host. Farmers can also generate a QR code with a link to their farm's page, which they can incorporate into their advertisements and product labels. This platform makes it easy for farmers to promote their farms and connect with potential customers.

Multilingual description assistance

The platform is planned to integrate a Large Language Model (LLM) that will analyze the location, activities, and products offered by the farm to generate descriptions based on existing online information. This feature is designed to allow farmers to seamlessly transfer information about their farm to the platform by providing a link to their website or other sources. Additionally, it will automatically generate multilingual translations, enabling farmers to expand their reach to a broader, global audience and significantly enhance their visibility.

Search system for visitors

The search system for visitors is an important component of the application. The platform is envisioned to allow visitors to quickly find their visit based on specific criteria. Designed to be user-friendly and intuitive, the system will allow users to search for farms by location, activities offered, and availability. This planned feature aims to help visitors identify the ideal farm for their needs.

Job listings

The proposed application is planned to include an integrated job board feature, enabling farmers to post both seasonal and permanent job openings. This functionality will allow farmers to advertise positions and attract potential candidates efficiently. Jobseekers will be able to apply directly through the platform, offering a streamlined and convenient solution for connecting with employment opportunities in the agricultural industry.

Review and rating system

The platform is designed to include a review and rating system that enables both farmers and visitors to provide feedback on their experiences. Farmers will be able to receive input from customers regarding the quality of their products and services, while visitors can rate their time spent at the farm. This planned feature aims to increase transparency, trust, and quality assurance between farmers and their customers and visitor confidence in their choices.

Networking and resource sharing for farmers

A planned feature of the application is its internal social network, designed to enable farmers to connect with one another, exchange insights, seek advice, and even coordinate equipment lending or borrowing. This feature will foster knowledge sharing in addition to collaboration between farmers and keep them updated and connected with their peers in the agritourism and agriculture industries.

Feedback portal for government

A planned unique feature allows the aggregation and anonymization of data, which can then be shared with governmental bodies. This provides valuable insights directly from the agricultural community.

4.2 Use-case diagrams

Use case diagrams are a component of UML that represent the functional requirements of a system. They represent interactions between actors (users or external systems) and the system itself, capturing high-level functionality and user goals. Such diagrams are required very early on during the system development life cycle, especially during the requirements analysis and design phases.

In system proposals, use case diagrams are commonly employed to convey the scope and purpose of the system to stakeholders so they can have a common understanding of user needs and system capabilities. They describe vital functionalities and user interactions and provide the basis for further detailed design, development, and testing to implement features that meet user expectations and business goals.

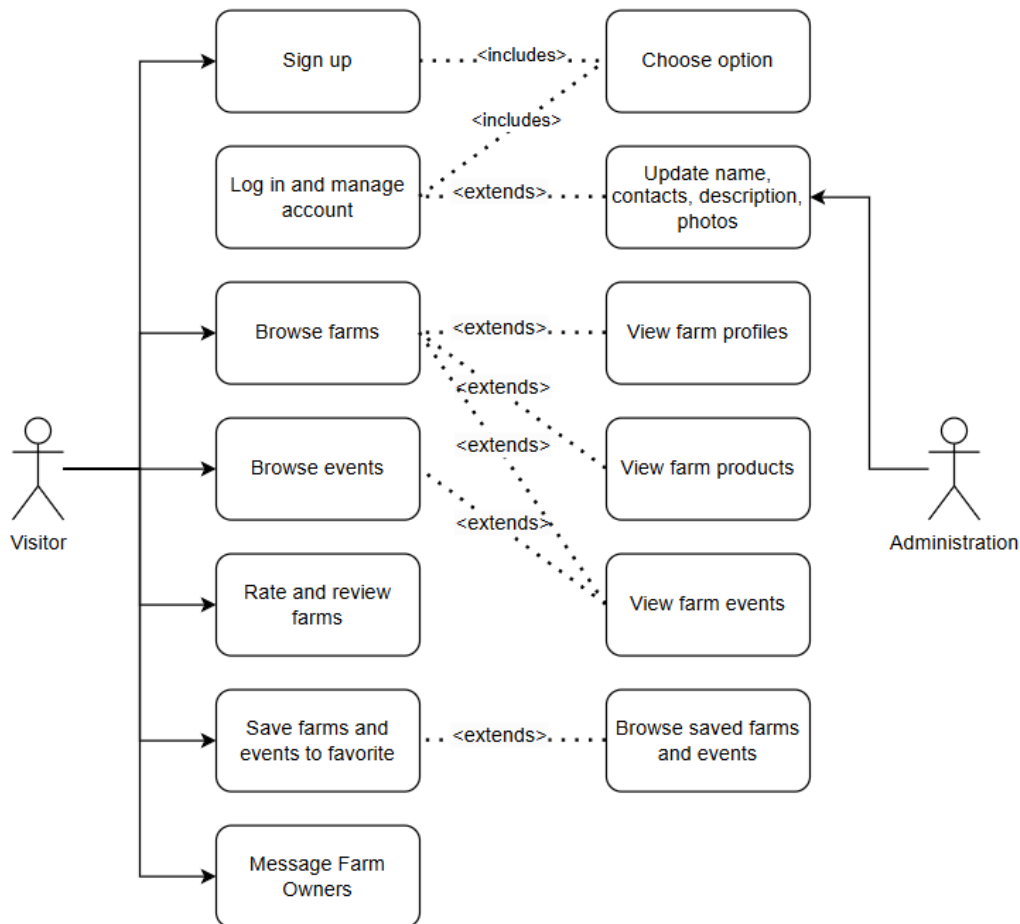


Image 7: Use case diagram visitor and administration (source: author)

Based on an analysis of user needs and agritourism system functionalities, the use case diagram illustrates the main roles and interactions within the application. The "Visitor" represents the general users of the platform, and the "Administration" handles system management.

The "Visitor" can register and manage their profile, browse farms and events, rate and review experiences, and save favorites. Each of these actions leads to more detailed views, including farm profiles, products, and events. The system also allows visitors to contact farm owners directly through automatically generated emails without having to leave the platform. The "Administration" role includes system management tasks, such as assisting users in updating their profiles when necessary.

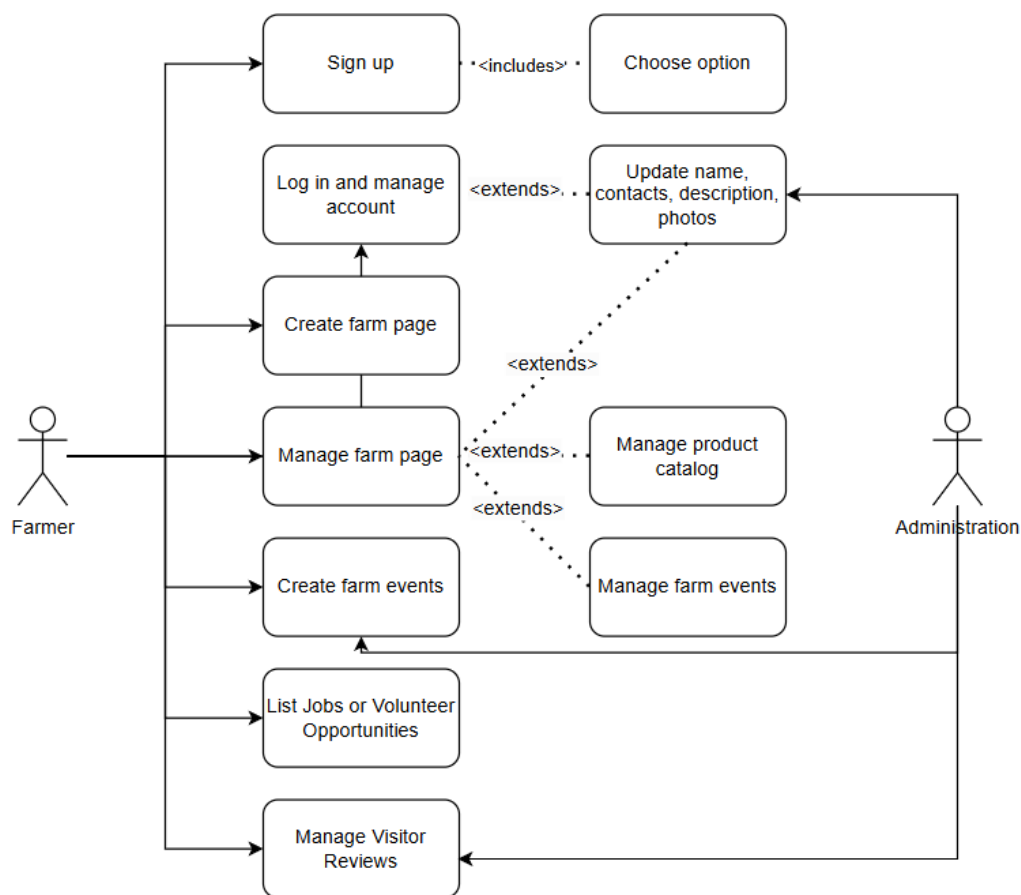


Image 8: Use case diagram farmer and administration (source: author)

Farmers' login and signup process is similar to that of visitors. Additionally, they can create and manage content about their farms. The "Create Farm Page" feature allows them to tell the story of their Farm and highlight services. They can also manage their product catalog and create seasonal events or activities via the "Create Farm Events" and "Manage Farm

Events" functionalities. Farmers can also view and reply to visitor reviews. The "Administrator" supports these interactions by checking that they follow platform guidelines and by offering help when required.

4.3 Wireframes

Wireframes are a simplified visual guide that represents the skeletal framework of a website. Prior to the design, wireframes are usually presented in black and white and do not define any visual style, typography, or imagery.

Wireframes focus on:

- The range of functions available.
- The relative priorities of the information and functions.
- The rules for displaying certain kinds of information.
- The effect of different scenarios on the display.

In a traditional design process, wireframes come after on-the-fly hand-drawn sketches and right before high-fidelity mockups or prototypes. There are three levels of wireframing, low fidelity, mid-fidelity, and high fidelity — though you can go straight from a low fidelity wireframe to a prototype and skip high fidelity wireframing as a distinct step. (Toni Gemayel, 2024)

Low fidelity is the most basic type of wireframing. They are done in grayscale with a focus on layout and high-level interactions. UI elements and content are represented by basic shapes like squares, triangles, circles, and lines.

Mid-fidelity wireframes help designers iterate and shape the final design. Design teams may add annotations and content as they try out different approaches to user flows and UI design elements, mapping out core functionality and key interactions. This enables teams to settle on the final wireframe design framework before adding visual design components.

High-fidelity wireframing takes what was created in the low-fidelity stage and sprinkles in more of the detailed elements. High-fidelity wireframes include visual markers and branding signifiers like colors, graphics, and font style. UI elements look realistic and might even include textures and shadows.

The website wireframe connects the underlying conceptual structure, or information architecture, to the surface or visual design of the website. In web design, wireframes are visual guides that represent the skeletal framework of a page or app screen. They commonly depict functional layouts, including interface elements and navigational systems. The wireframe usually lacks typographic style, color, or graphics since the main focus is on functionality, behavior, and priority of content. This is a direct consequence of their CAD history. In other words, it focuses on what a screen does, not what it looks like. Wireframing usually takes place at the beginning of a web design project. In this phase, designers define the project

requirements, exchange ideas, and test the scope of the project before actually starting the design phase. (Figma, 2024)

Figma is one of the most versatile platforms for creating interactive wireframes and designing mockups and prototypes in the browser or desktop. It supports collaboration, cloud storage, component team libraries, and many integrations. Thus, it was chosen for the development of wireframes. The fidelity rate was low to medium, starting with low in order to map out the general layout and increasing the number of details in future page wireframes to bring more identity and design.

4.3.1 Home page

The main screen of the application is what users see first when visiting the website and is designed to allow users to browse and interact with content easily. The key components of this interface include:

- **Header:** Located at the top of the page, the header provides constant access to essential options via a profile dropdown menu and prominently displays the application logo for consistent branding and easy navigation. It ensures that users can quickly access their account settings or return to the homepage by clicking the logo from anywhere in the application.
- **Navigation Pane:** Located on the left side of the screen, it includes buttons for accessing various sections of the application, such as "Home," "Events," "Jobs," "Farms," and "Saved."
- **Map Section:** A prominent map occupies the top central part of the screen, serving as a visual tool for locating farms and events.
- **Farm Cards:** Below the header, individual farm cards are displayed in a grid format, each containing an image placeholder, key details, and descriptive text about the farm or activity. The grid layout ensures a clear and organized presentation for users to browse multiple farms at a glance.
- **Footer:** Positioned at the bottom of the page, the footer provides additional navigation options or relevant links to enhance user experience.

The wireframe highlights the arrangement of these components, ensuring the interface is intuitive and visually appealing. This visual design serves as the foundation for implementing future concepts and a functional front-end for the application.

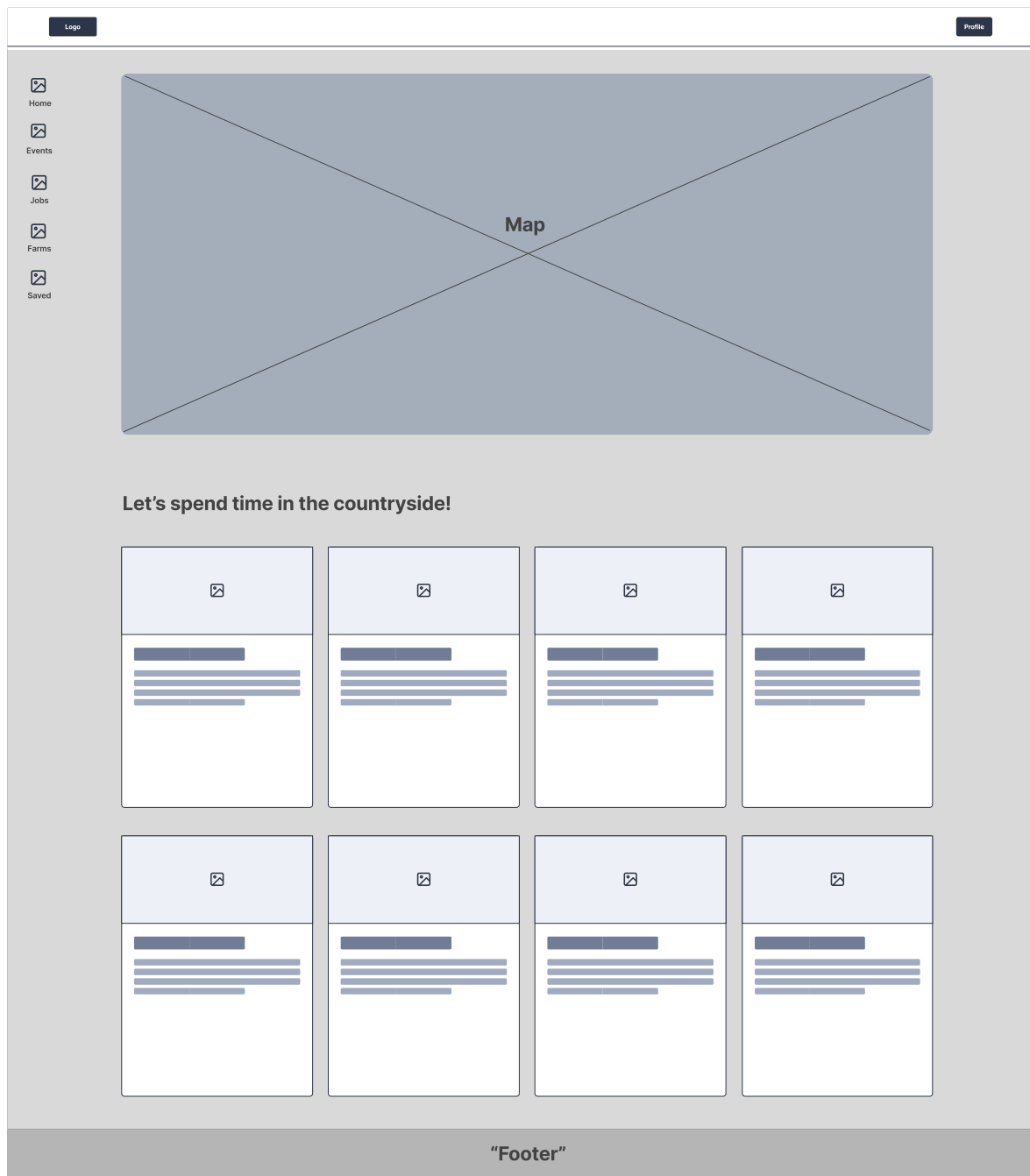


Image 9: Home page wireframe (source: author)

4.3.2 Farm page

The farm page was designed with mid-fidelity to evolve and capture the website's identity and design language. This wireframe represents the farm detail page in the proposed application, designed to provide visitors with detailed information about a specific farm and its offerings. The layout includes standard elements described previously such as the header and navigation panel, ensuring consistency across the application. The key components of this interface include:

- **Language selector:** Users can switch languages using a dropdown option, ensuring inclusivity for a global audience.
- **Farm Information:** The farm's name, location, owner details, and a star-based rating system are prominently displayed, along with options to edit (displayed only for the owners of the current farm page), share, or save the farm profile.
- **Image Gallery:** A visual section featuring a grid layout to showcase the farm through multiple photographs, helping users get a visual understanding of farm offers.
- **Farm Description:** A section for farmers to provide narrative covering unique features, sustainability practices, and standout offerings such as signature products or special attractions.
- **Amenities and Activities:** A section listing the available amenities and experiences (e.g., "Wine Degustation," "Farm Yoga with Goats") with a "Show All Amenities" button if there are more to be displayed.
- **Farmer Contact Card:** a section displaying essential contact details, a profile image, and a QR code that could be used to share this current farm page and a contact form.

This page prioritizes visual appeal, information density, and interactive elements, to provide a seamless experience for visitors exploring farm profiles.

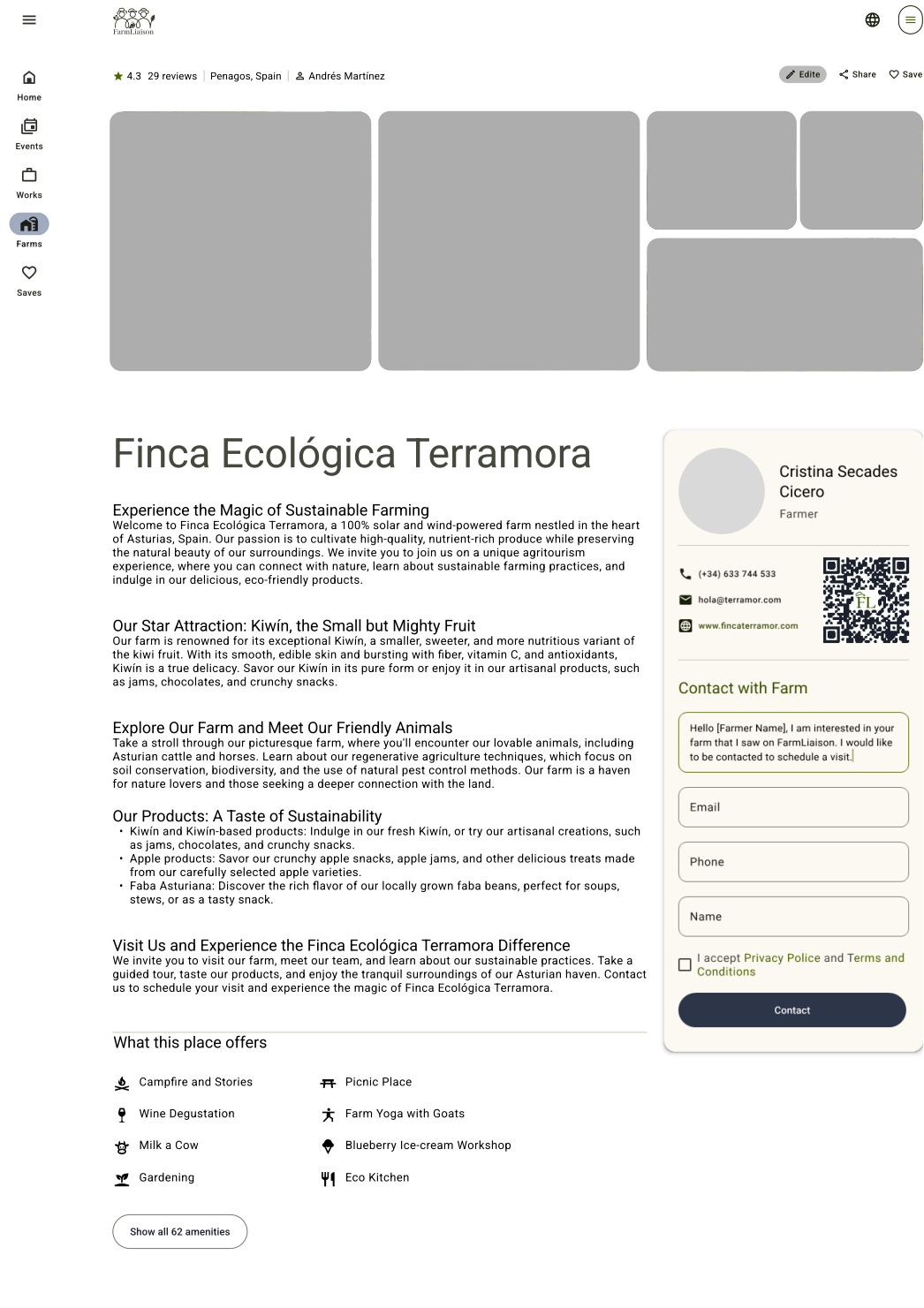


Image 10: Farm page wireframe (source: author)

4.4 Website Prototypes

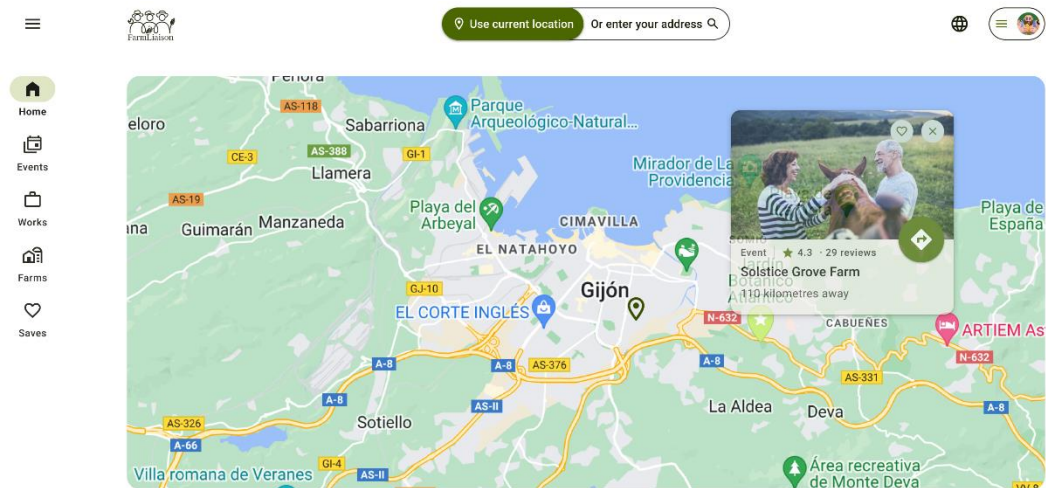
The next step was to design high-fidelity website prototypes to create a more detailed and realistic visualization of the proposed application. These prototypes aim to simulate the final product, showcasing the layout, design elements, and functionality with greater precision. The goal is to refine the user interface for a better understanding of the application design language.

This chapter presents high-fidelity prototypes for key pages of the application, focusing on visual consistency, usability, and functionality. Each design iteration incorporates insights from prior research and wireframe designs to ensure consistency. The prototypes are intended to provide a clear representation of how users will perceive the interface.

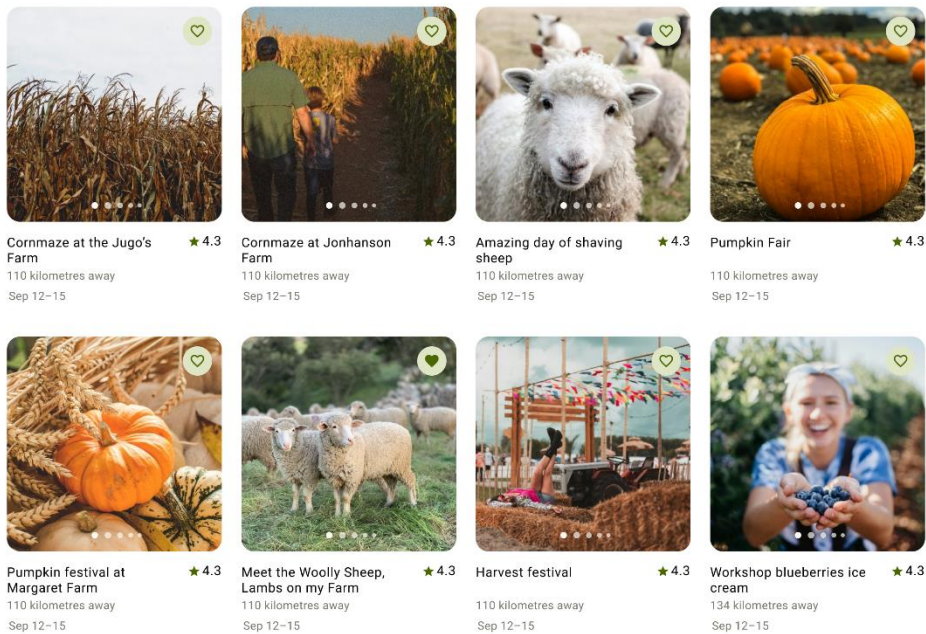
4.4.1 Home page

Significant updates and refinements to the home page since the wireframe include:

- **Enhanced visual design:** The addition of color schemes, icons, and typography. Farm and event images are now integrated into the layout, replacing placeholders, to create a better representation of final application.
- **Dynamic cards:** Cards now represent the ability to scroll through preview pictures, add farm or event to favorites using the heart icon in the top right corner.
- **Load more option:** A "Show more" button has been introduced at the bottom of the event list to let users to explore additional events dynamically, without overwhelming the initial view.



Let's spend time in the countryside!



Continue exploring events

Show more

Image 11: Home page prototype (source: author)

4.4.2 Farm page

Since the farm page was initially developed in mid-fidelity this advanced prototype brings in better visual representation along with additional components accessed by scrolling down: : Farm page part 2 (source: author) and : Farm page part 3 (source: author).

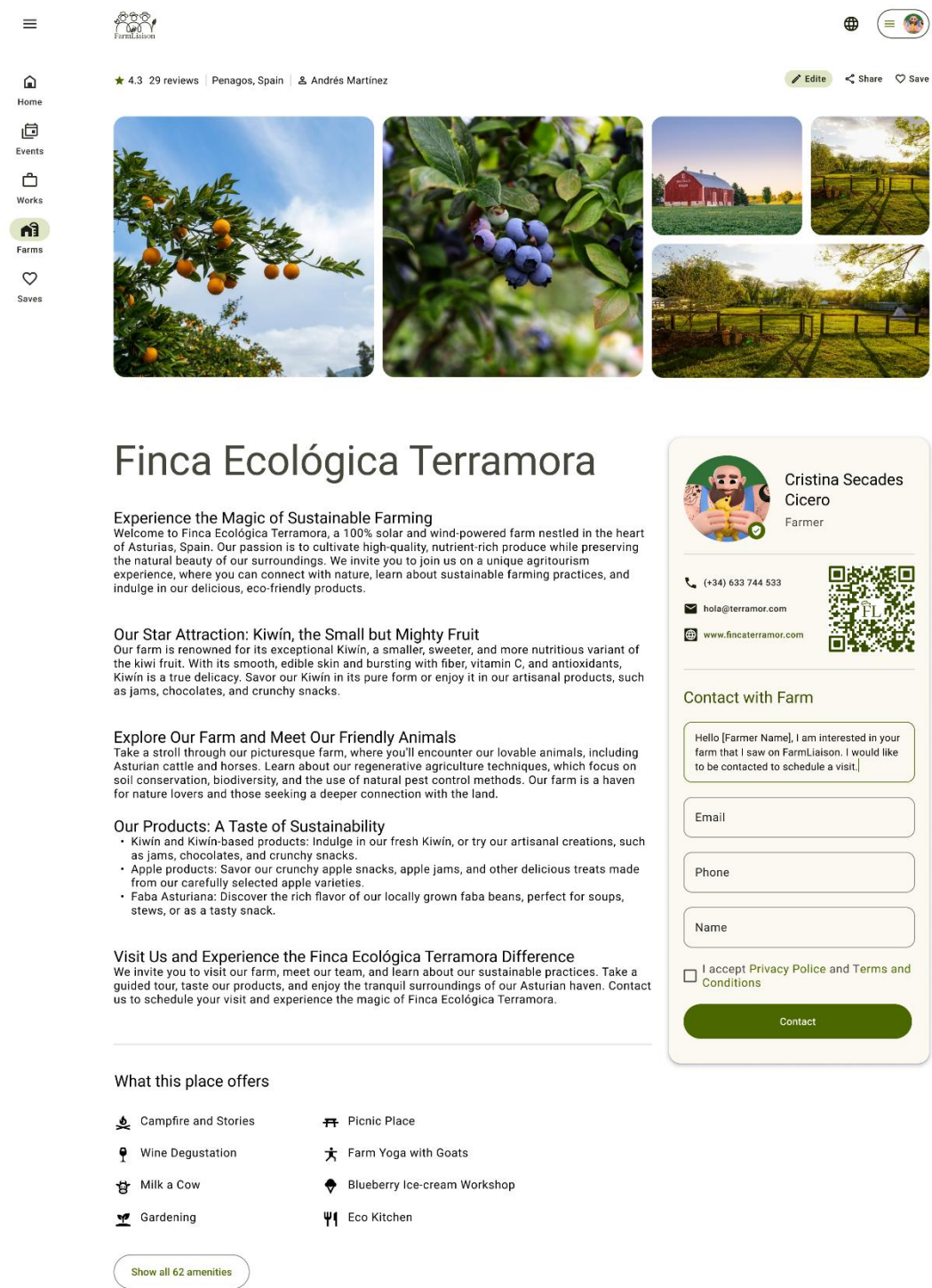


Image 12: Farm page part 1 (source: author)

This section represents available farm offerings, such as products and events (if those exist). A similar card design is used to represent information in separated blocks.

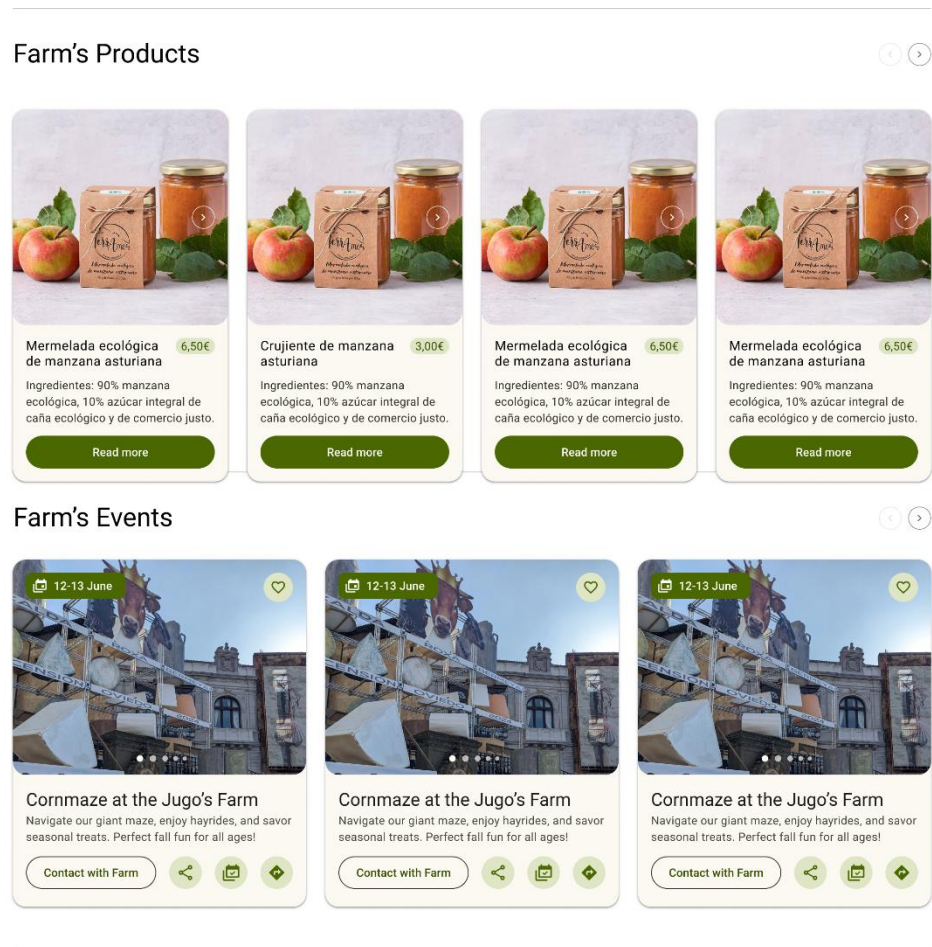


Image 13: Farm page part 2 (source: author)

The further section Is designed to provide space for users to leave reviews and view already existent ones, with an additional interactive map at the bottom of the page to provide users with a better understanding of the farm location.

Share Your Experience

We'd Love to Hear About Your Time with Us!

 Anke



Share details of your own experience at this event

2000 characters

CancelPost

 Anke
August 2023

I recently visited a charming farm Rioja Vineyard Retreat, nestled in picturesque countryside. The stunning views, hands-on farm activities, and farm-to-table dining were highlights. The warm hospitality and commitment to preserving Spanish traditions made this retreat truly special. Highly recommended for a tranquil and authentic Spanish experience.

 Raymond
August 2023

My recent visit to Sierra Sunrise Ranch was a delightful escape. Surrounded by stunning countryside views, I enjoyed hands-on farm activities and savored delicious, locally sourced meals. The warm hospitality and commitment to sustainability made it a truly memorable experience. Highly recommended for a taste of rural Spain.

Share review

Report review

Spanish farm left much to be desired. The countryside setting was lovely, but the accommodations were rather basic and lacked modern amenities. I found the farm activities to be more labor-intensive than enjoyable, and the limited variety of activities left me feeling somewhat bored during my stay.

Where you'll be

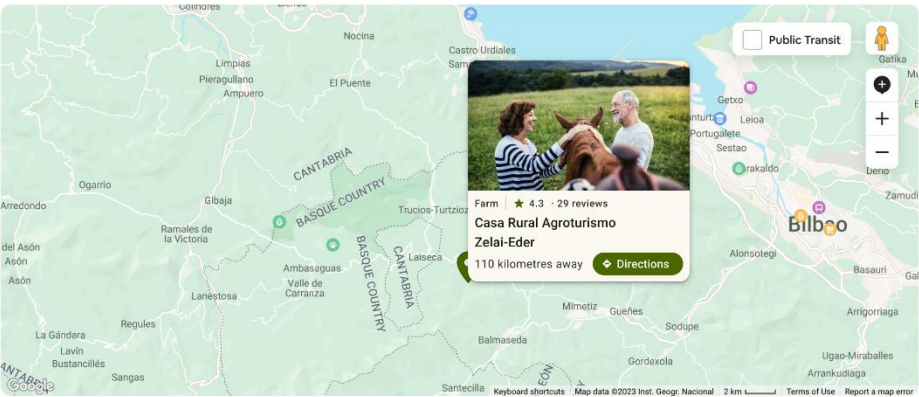


Image 14: Farm page part 3 (source: author)

4.4.3 Event page

The event page follows a similar layout to a farm page for consistency purposes, with additional elements displaying the date and the farm at which the event is hosted.

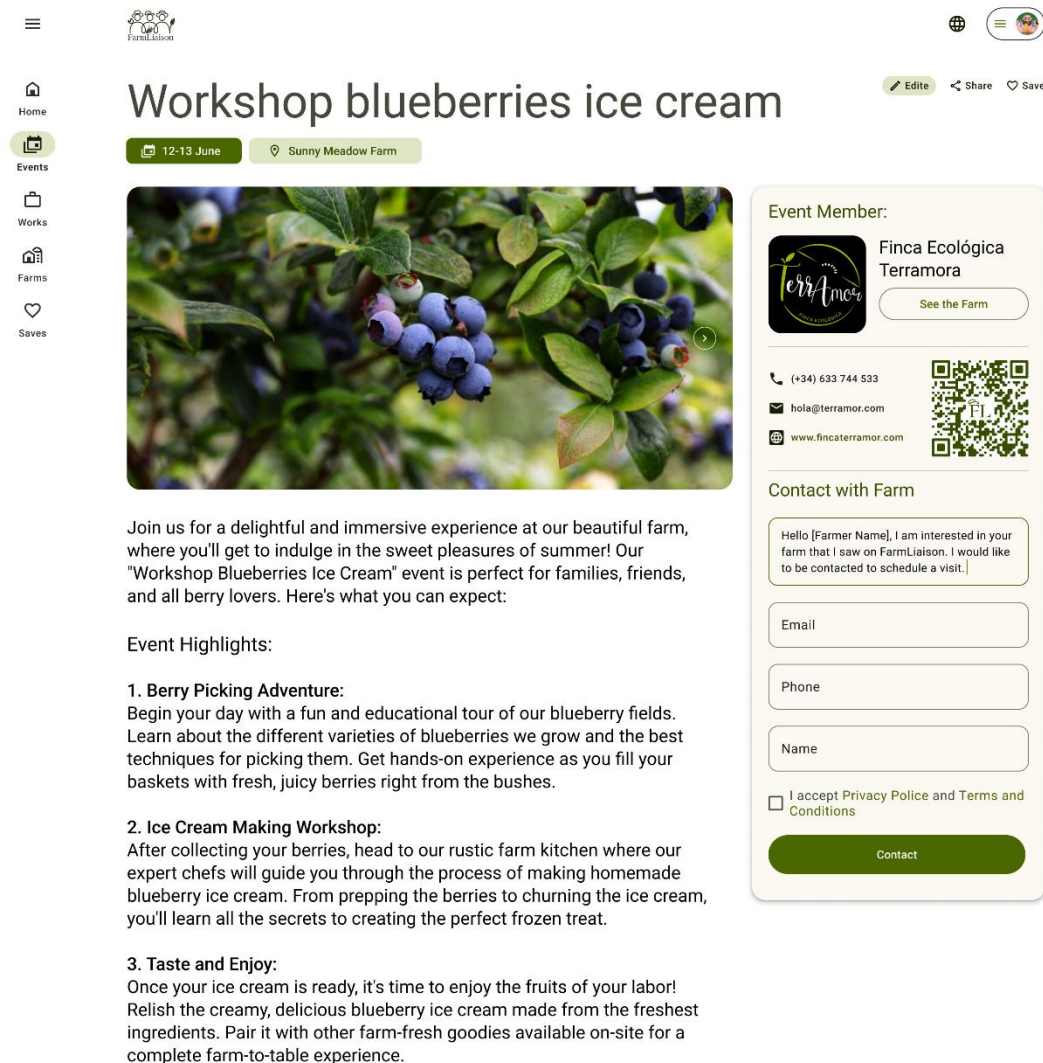


Image 15: Event page (source: author)

4.4.4 Profile pages

The default profile page (Image 16) is designed to provide a comprehensive overview of the user's activities and interactions on the platform. It focuses on personalization, user interaction history, and review sharing. Key components of the user profile page:

Profile overview:

- **Profile picture and name:** At the top, the user's profile picture and full name are displayed.
- **Role and location:** The user's role on the platform and their location are clearly indicated, along with their preferred languages.
- **Edit profile button:** A "Edit Profile" button allows users to update their profile information conveniently.

- **About section:** This section contains a short, personalized bio written by the user detailing their interests and relationship with agritourism.
- **Visited farms section:** Introduces a showcase of farm cards the user has visited.
- **Reviews section:** A detailed paragraph of feedback about the user's experience with a specific farm.

The profile page is a user-centric hub, allowing individuals to manage their presence on the platform, revisit past experiences, and share feedback.

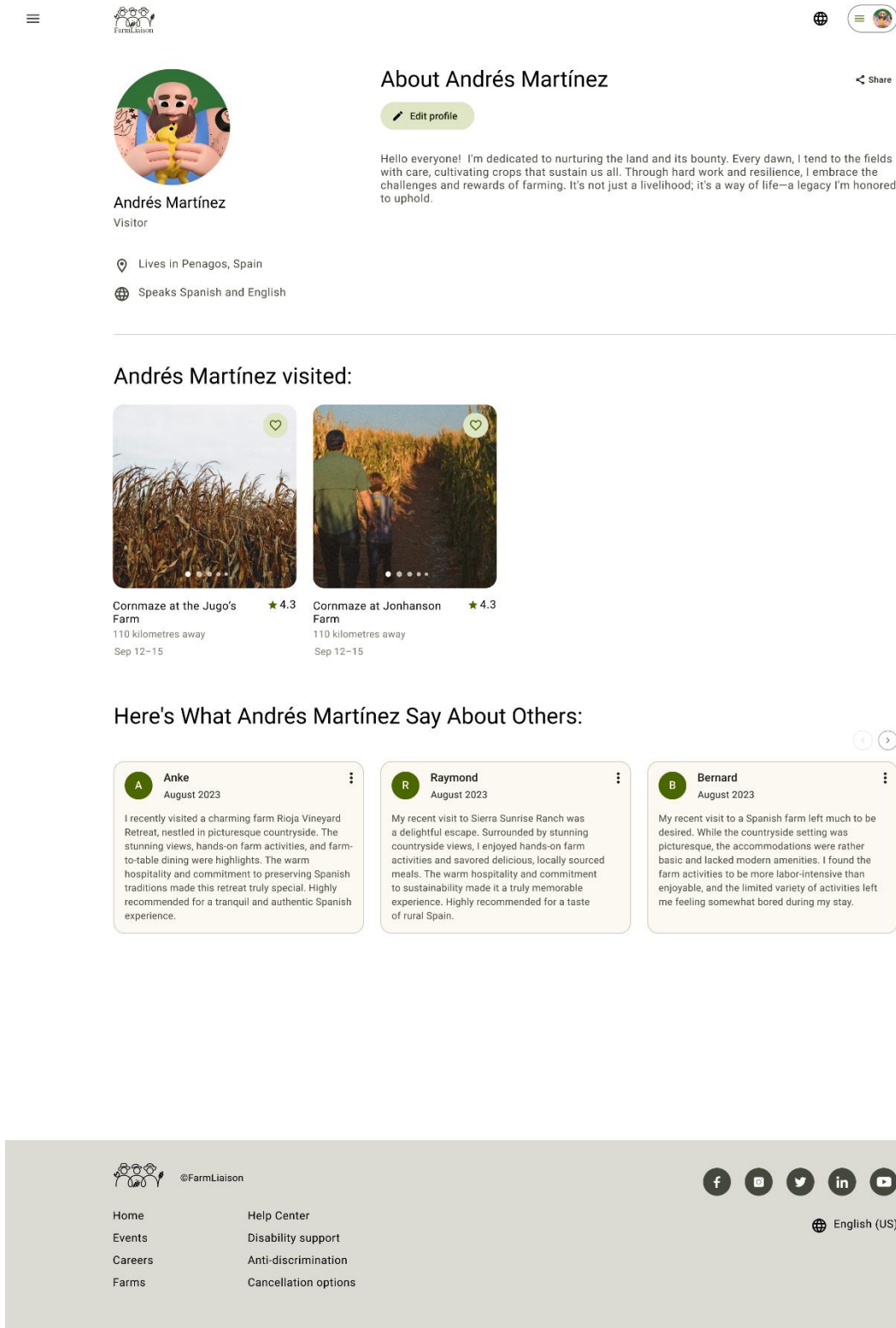


Image 16: Profile page (source: author)

Editing the profile page contains a list of text inputs so that users can update their information like name, email address, phone number, and description.



Your profile

The information you share will be used across FarmLiaison to help other guests and Farmers get to know you.

My name*

My email address*

Use an address you'll always have access to.

My phone number*

 Phone number

Where I live:

Languages I speak:

About me:

Write something fun and punchy.

122 characters

I am:

☒ Visitor ☐ Farmer ☐ Worker



Image 17: Editing profile page (source: author)

5 Development

This chapter examines the application structure by detailing the directory tree and providing brief descriptions of individual directories. It explores the specific front-end frameworks used in the proposed application, including Vue.js, Nuxt 3, and Quasar, describing how they interact with each other and support the platform's development. These frameworks were essential to the application's architecture and contributed significantly to its functionality and performance.

The chapter also analyzes additional web technologies that were studied during development, such as SSR rendering, component-based architecture, asynchronous loading, lazy loading, and others. While not all of these technologies were fully integrated into the proposed application, they provided valuable insights into modern web development practices and influenced some design and implementation decisions.

Furthermore, this chapter delves into the database structure and its interaction with the front end. A deep understanding of the database schema and the endpoints for accessing data was crucial in designing the front end, as the interface relies heavily on retrieving, displaying, and interacting with this data. Finally, real code examples are provided to illustrate how some of these frameworks and technologies were implemented in the application.

5.1 Application structure

The application structure defines how the project's files and directories are organized to provide a logical and efficient workflow for development and maintenance. A clear structure is important for any web application because it helps developers navigate the codebase, understand its functionality, and collaborate effectively.

The directory structure follows a standardized organization commonly used in modern web development, particularly with Nuxt.js. Each directory serves a specific purpose, such as managing the user interface, handling application logic, or storing configuration files.

Code 1: Directory structure (source: author)

```
.
├── .github
├── .nuxt
├── .vscode
├── components
├── composables
├── css
├── layouts
├── locales
├── middleware
├── node_modules
├── pages
├── plugins
├── prisma
├── public
├── server
├── types
├── utils
├── .env
├── .gitignore
├── .nuxtignore
├── app.vue
├── nuxt.config.ts
├── package.json
├── README.md
├── tsconfig.json
└── yarn.lock
```

5.1.1 Directories descriptions

This section describes some of the key directories and ones directly correlated to the front-end part of the application, excluding ones like GitHub-related directories, configuration directories, server, and database-related directories in order to maintain relevance to the scope of this thesis.

- **Components directory**

The components directory contains reusable Vue.js components, which serve as the foundational building blocks for the user interface. These components are designed to handle specific tasks, making the application modular, maintainable, and efficient. The directory is organized to include various UI elements that can be reused across multiple pages and sections of the platform.

- **CSS directory**

The CSS directory contains stylesheets that define the visual appearance and design of the application. It includes global styles and specific configurations so that they do not have to be configured multiple times.

- **Layouts directory**

The `layouts` directory contains reusable Vue.js layout components that define the overall structure of various pages within the application. These layouts serve as templates, organizing common elements like headers, footers, and side navigation bars to provide consistency across multiple pages.

The `default.vue` layout functions as the primary template for most pages, incorporating a header, footer, and navigation bar. The `error.vue` layout is specifically designed for error pages, such as 404 errors or server-related issues. The `limited.vue` layout is used for pages where content needs to remain within a confined width, optimizing readability and usability, especially on wider monitors.

- **Locales directory**

The `locales` directory contains JSON files that store the application's translation strings, enabling support for multiple languages. Each file corresponds to a specific locale, such as `en-US.json` for English (United States), `es-ES.json` for Spanish (Spain), and others like `fr-FR`, `de-DE`, and `it-IT`. These files allow the application to switch between languages based on user preferences or browser settings.

The structure of each file includes key-value pairs, where the keys represent identifiers for text elements in the application, and the values contain the translated text for the respective language.

- **Pages directory**

The `pages` directory contains Vue.js files that define the structure and content of individual pages in the application. Each file in this directory corresponds to a specific route in the application, so that Nuxt can automatically generate the routing system. Subdirectories within `pages` represent nested routes for a hierarchical organization of the application's navigation.

For example, the `index.vue` file serves as the default homepage, while other files like `about.vue`, `contact-us.vue`, and `faqs.vue` represent standalone pages with specific content. Subdirectories like `event` and `farm` contain route-specific components for events or farms, allowing dynamic rendering of detailed information.

This directory is used for defining the application's routing and structuring its content in a clear, modular way.

- **Public directory**

The `public` directory contains static assets such as images, icons, documents and other files that do not change frequently.

Key examples include a `favicon.ico` file for the website's favicon and localized HTML files like `terms-en.html` or `terms-de.html`, to provide language-specific terms and conditions.

The `img` subdirectory contains static images used across the application, such as logos, banners, or icons.

5.2 Front-end frameworks (Vue.js, NUXT 3, Quasar)

In this section the utilized tech stack is partially described, mainly front-end frameworks and what features differentiate them. In web development they provide a structured foundation of pre-written code and tools to streamline the development process, making it easier to build complex applications with consistent architecture and reusable components. The architecture of almost all frameworks is based on the decomposition of several separate layers (applications, modules, etc.) of a project. This means that it is possible to extend the functionality of the application based on the needs and use the modified version together with the framework code or leverage third-party applications. This flexibility is one of the key advantages of using frameworks. Quasar, NUXT 3, and Vue.js build upon each other to provide a layered approach to development, each adding specific functionalities to streamline the process.

5.2.1 Vue.js

Vue.js is a progressive JavaScript framework for building user interfaces of any complexity, designed to be flexible and incrementally adaptable. It builds on top of standard HTML, CSS, and JavaScript and provides a declarative, component-based programming model that helps efficiently develop user interfaces of any complexity. Two of the core features of Vue are:

Declarative Rendering: Vue extends standard HTML with a template syntax that allows to declaratively describe HTML output based on JavaScript state.

Reactivity: Vue automatically tracks JavaScript state changes and efficiently updates the DOM when changes happen. (Vue.js, 2024)

5.2.2 NUXT 3

NUXT 3 is a framework built on top of Vue.js, aimed to create type-safe, performant and production-grade full-stack web applications and websites. It provides advanced features like server-side rendering (SSR) and automatic routing, along with middleware support, and a structured file-based approach to organizing projects. NUXT 3 is designed to enhance performance, improve SEO, and streamline the setup of universal applications by handling complex configurations out of the box. (Nuxt, 2024)

5.2.3 Quasar

Quasar is a UI framework built on top of Vue.js (and compatible with NUXT 3) that provides a collection of pre-built, responsive UI components. It supports cross-platform development, allowing the creation of applications that can be deployed as web, mobile, or desktop applications with minimal configuration changes. Quasar is known for its extensive library

of components, theme customization options, and plugins, which help create cohesive, visually consistent interfaces across devices. (Quasar, 2024)

While Quasar's UI components are visually compelling and provide a ready-to-use foundation, additional CSS adjustments are made to create a unique visual identity for the application.

5.3 Modern web technologies in practice

It is important to understand how some modern web technologies function while developing the proposed application, as they significantly influence its performance, scalability, and user experience. This section examines a selection of web technologies that were analyzed for their potential to meet specific development needs, such as building dynamic user interfaces, optimizing performance with server-side rendering, and ensuring responsive design across devices. Although some of these technologies were not directly implemented in the application, their analysis offered critical insights that shaped important development choices.

5.3.1 Component based architecture

Component based architecture in web development focuses on breaking an application into small, reusable parts called components. Each component in frameworks like **Vue.js** and **Nuxt 3** contains its own logic, template, and style, making it easier to develop, test, and maintain. In this setup, components are responsible for managing dynamic data and ensuring the user interface reflects any changes efficiently. By using Nuxt 3 with Quasar Framework, components can be optimized for better load times and performance. This modular approach simplifies collaboration between the front-end and back-end systems, like Prisma ORM and PostgreSQL, by clearly defining how data is passed and rendered. (Vue.js, 2024)

5.3.2 Server-Side Rendering (SSR)

Server-Side Rendering (SSR) is a technique where HTML for a web page is generated on the server and sent to the client as a fully rendered page. Unlike client-side rendering, where the browser uses JavaScript to build content dynamically, SSR provides the user with ready-to-display content immediately. This approach improves the perceived loading speed of a page, as the browser does not need to wait for JavaScript to load and execute before displaying the content. (Vue.js, 2024)

Once the initial HTML is loaded by the browser, hydration occurs. Hydration is the process by which client-side JavaScript "activates" the static HTML sent from the server, attaching the JavaScript logic needed for interactivity. This step allows the SSR-generated HTML to become a fully interactive single-page application (SPA), handling user inputs and updates seamlessly. (Vue.js, 2024)

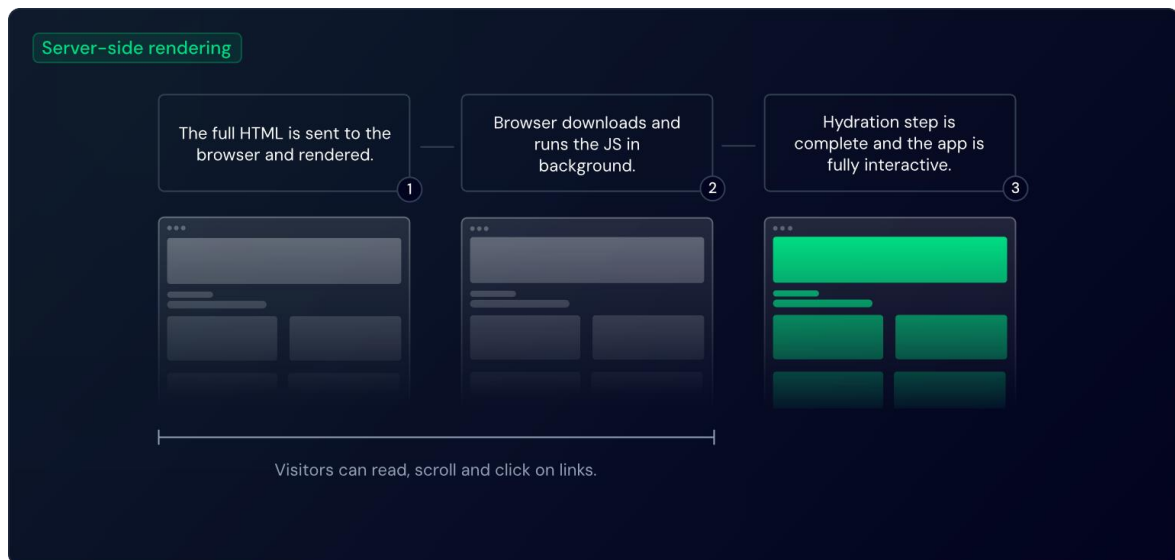


Image 18: SSR Diagram (source: Nuxt, retrieved October 30, 2024)

SSR brings several advantages, particularly in terms of performance and accessibility. By delivering a pre-rendered page, SSR reduces the amount of JavaScript that needs to be processed on the client, which is beneficial for users on low-powered devices or slower networks. Additionally, SSR enhances search engine optimization (SEO) since search engines can index the page's content immediately. This technique also improves accessibility, making content available to screen readers and assistive technologies from the start without relying on client-side rendering. Furthermore, SSR enables easier caching of content on the server side, which can speed up repeat visits by reducing server load and response times. (Nuxt, 2024)

5.3.3 Client-side Routing

Client-side routing refers to the process where HTML content is dynamically generated by JavaScript in the browser, enabling faster page updates. When users interact with the application, changes occur almost instantly, enhancing the user experience by eliminating delays typically caused by server requests for new content.

With client-side routing, the determination of which View to display based on the URL happens in the browser through JavaScript. Unlike server-side routing, where the server decides which resource to send, client-side routing involves the server returning a single HTML file. The JavaScript in this file then dynamically determines the content to display by interpreting the URI. In this context, "routing" refers to the process of mapping a URI to the corresponding View or representation of a resource. A "route" is a specific URI that corresponds to a particular View. (Joel Ross, 2024)

5.3.4 Progressive web application (PWA)

Progressive Web Applications (PWAs) are a modern web development technology that transforms websites to function more like native applications. Built using HTML, CSS, and

JavaScript, PWAs are hosted on web servers and executed in web browsers. They can be accessed via web browsers like a traditional website, installed directly from a supporting browser, or distributed through app stores.

PWAs support various architectural styles, including Client-Side Rendering (CSR), Server-Side Rendering (SSR), or a combination of both. This flexibility allows developers to leverage the strengths of both approaches, resulting in enhanced responsiveness and performance. For example, PWAs are particularly advantageous for E-commerce websites due to their ability to offer a seamless and engaging user experience. (Elfström Adam, 2021)

Key Benefits of PWAs:

- Search Engine Optimization (SEO): PWAs can be indexed by search engines, making them discoverable like traditional websites.
- Ease of Distribution: PWAs can be accessed and launched using a standard web link.
- Security: They use HTTPS endpoints and modern web security measures, ensuring a safe user experience.
- Responsiveness: PWAs adapt dynamically to various screen sizes, orientations, and input methods.

Adopting PWA technologies is a powerful way to create secure, discoverable, easily installable, and responsive apps. Notable successes demonstrate the value of PWAs. For instance, Trivago reported a 150% increase in users adding their PWA to home screens, leading to a 97% rise in hotel offer visits. (Trivago, 2017)

5.4 Database structure

The database serves as the central repository for all data, including user accounts, farm profiles, products, jobs, and activities. It ensures data integrity, supports relationships between entities (e.g., farms and products), and provides the means to retrieve or update records based on user actions.

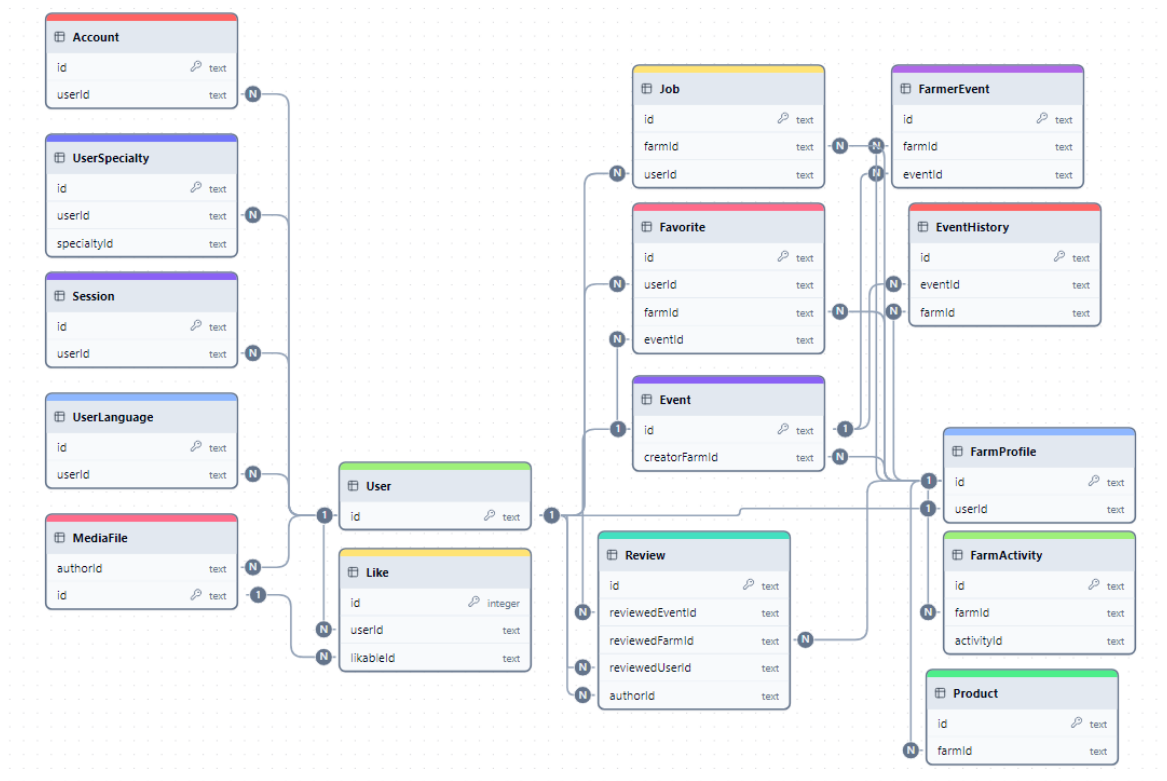


Image 19: E/R database diagram (source: author)¹

The database utilizes PostgreSQL with Prisma, with the rationale for selection being:

Reliability and Performance: PostgreSQL is noted for its robustness and ability to deal with huge volumes of information without compromising performance. With the proposed application aiming to onboard numerous farms and users, this reliability is important.

Complex Features: PostgreSQL provides a plethora of skilled features such as table partitioning, Point-In-Time Recovery, and much more. This is useful as the platform evolves and its database requirements start to be much more advanced.

Prisma: Using Prisma with PostgreSQL makes the database much more type-safe. Moreover, the autogenerated query builder ensures development speed and runtime safety. This pairing means a shorter time spent on debugging and much more on feature delivery.

Flexibility: As the application grows, there may be a necessity for real-time features, analytics, or maybe complicated queries. PostgreSQL, mixed with Prisma, provides the flexibility to focus on these requirements.

¹ Due to the complexity of database the diagram was slightly simplified, the full interactive version could be viewed at: <https://dbdiagram.io/d/67328221e9daa85aca164a45>

5.5 Front-end interaction with APIs and database

In web applications, the front end does not directly interact with the database. Instead, it communicates with the back-end through APIs, which act as a bridge between the two layers. This indirect interaction ensures that the front end focuses on user experience while the backend handles data processing and storage securely. In the web application, the front end accesses the database indirectly by making HTTP requests to API endpoints using the ``$fetch`` function. These requests are processed by the back-end server, which interacts with the database to perform the necessary operations, such as retrieving or updating data. The following subchapters describe some of the main functions developed and how they communicate with the back end.

5.5.1 Fetching Farm Data

Code 2: Example of the code retrieving information about farm (source: author)

```
1. const fetchFarmData = async () => {
2.   try {
3.     const farmData = await $fetch(
4.       `/api/${locale}/farm/${route.params.farmSlug}`,
5.     );
6.
7.     if (farmData) {
8.       const transformedActivities =
9.         farmData.activities?.map((activity) => ({
10.           value: activity.id,
11.           label: activity.name,
12.         })) || [];
13.
14.       farmData.activities = transformedActivities;
15.       editedFarm.value = { ...toRaw(farmData) };
16.     } else {
17.       console.error('Farm data is missing:', farmData);
18.     }
19.   } catch (error) {
20.     console.error('Error fetching farm data:', error);
21.   }
22. };
23.
```

The function `fetchFarmData()` retrieves farm details from an API endpoint (`/api/${locale}/farm/${route.params.farmSlug}`) and updates the `editedFarm` object with the response.

- This code snippet demonstrates a **GET** request sent from the front-end to fetch farm-related data from an API endpoint (`/api/${locale}/farm/${route.params.farmSlug}`).
- The response is transformed to fit the UI model (e.g., mapping activity IDs and names).
- The data is then assigned to the `editedFarm` object, enabling dynamic rendering of the page.

5.5.2 Updating Farm Data

Code 3: Example of the code updating information about farm (source: author)

```

1. const updateFarm = async () => {
2.   loading.value = true;
3.   editedFarm.value.translateAll = translateAll.value;
4.   try {
5.     await $fetch(`/api/${locale}/farm/${route.params.farmSlug}`, {
6.       method: 'PUT',
7.       headers: {
8.         'Content-Type': 'application/json',
9.       },
10.      body: editedFarm.value,
11.    });
12.    $q.notify({ type: 'positive', message: 'Content updated successfully' });
13.    router.push(localePath(`/farm/${route.params.farmSlug}`));
14.  } catch (error) {
15.    console.log('Error:', error);
16.  } finally {
17.    loading.value = false;
18.  }
19.};
20.
```

The `updateFarm()` function sends a **PUT** request to the back-end (`/api/${locale}/farm/${route.params.farmSlug}`) with the updated farm data in the request body.

- This example illustrates a **PUT** request used to send edited farm data to the back-end via the API endpoint.
- The body of the request contains the `editedFarm` object, which represents the updated state of the farm.
- Once the back-end processes this data and updates the database, the front-end notifies the user of success and redirects them.

5.5.3 Handling API Errors

Error handling ensures that any issues with the API (e.g., network errors or invalid responses) are logged for debugging. This mechanism improves robustness by preventing the app from crashing in case of unexpected API failures.

Code 4: Example of handling errors (source: author)

```
1. catch (error) {  
2.     console.error('Error fetching farm data:', error);  
3. }  
4.
```

5.5.4 User Inputs

Inputs from forms or user actions trigger functions like `updateFarm()` or `fetchFarmData()`.

Code 5: Example of the user input field to edit farm address (source: author)

```
1. <!-- Edit Address -->  
2.     <q-input v-model="editedFarm.address" label="Address" />  
3.
```

5.5.5 Dynamic Rendering

This snippet demonstrates dynamic rendering based on the presence of products in the `editedFarm.products` array. If products exist, they are displayed using a v-for loop. Each product is rendered with a reusable component (`CarouselProductCard`). If no products exist, a fallback message (No products added.) is shown.

Code 6: Example of implementing dynamic rendering (source: author)

```
1.     <div  
2.         v-if="editedFarm && editedFarm.products?.length > 0"  
3.         class="product-grid"  
4.     >  
5.         <div v-for="product in editedFarm.products" :key="product.id">  
6.             <CarouselProductCard :item="product":editable="isAuthor" />  
7.         </div>  
8.     </div>  
9.     <div v-else>  
10.         <p>No products added.</p>  
11.     </div>  
12.
```

6 User testing

The last, but important part of web application development is functional testing. This section is dedicated to verifying the correctness of the implementation according to the design. The goal of testing is to ensure the proper functioning of the application and overall user experience while. User testing was conducted with the participation of several testers, who subsequently provided their feedback.

6.1 Choice of features for user testing

It is necessary to decide which features are important and essential to the operation of the application. The following features are selected for testing:

- User registration, profile creation, and login
- Create a farm page
- Edit farm page
- Favorites and wish list
- Review and rating system
- Getting in touch with the farms

User registration and login:

Testing the user registration and login process ensures new users can create accounts and returning users can log into their profiles. This includes verifying the sign-up process and social media login options as well as completing steps to agree to terms and conditions and setup initial profile information.

Creating a farm page:

Testing the farm page creation feature to allow farm owners to set up their farm profiles. This includes adding the farm name, location, description, photos, and contact information.

Editing farm page:

Testing if the owners can update their pages with new photos, seasonal changes, and promotional details through editing functionality. A well-structured farm page setup process gives potential visitors accurate and up-to-date information about each farm.

Favorites and wish list:

Testing whether farms or activities could be saved to the favorites list for later reference. This helps with user engagement - by allowing visitors to track farms they like and return to them quickly.

Review and rating system:

Testing functionality for leaving reviews, rating farms, and reading feedback from other users. This helps users make informed decisions and build trust on the platform through user-sharing experiences.

Getting in touch with the farms:

Testing the "Contact the Farmer" feature means users can contact farm owners via a simple, on-page contact form. This allows users to enter their name, contact information, and message on the farm page. When the user presses "Send," an automated email with the user's inquiry is sent to the farm owner. This allows for instant communication without users leaving the website to send an email.

6.2 User testing method

During the development process, user testing was conducted remotely using the unsupervised method. Testers were presented with a task and expected output. The testing was recorded using screen and audio recording software for future reference. This will allow for a detailed analysis of the testing process, identify problem areas, and make adjustments in the further development process. All the testers' answers were documented using Google Forms.

For example, testers were given the task "create a farm page" and provided with a table for feedback on each step, indicating whether the web application functioned properly and whether the task step was successful, unsuccessful, or partially completed. Moreover, for detailed evaluation, an additional field was provided where testers could leave written feedback on whether their expectations regarding the functioning of the application were met or provide general recommendations for improvement.

For a more detailed analysis of the application's functioning, the author paid particular attention to how users interacted with the interface, including where they positioned their cursor while attempting to complete a step and how they perceived pop-up notifications. Additionally, the author monitored whether any aspect of the interface caused confusion or frustration for the users. During the testing, the author has not provided users with any specific guidelines on how to achieve certain tasks. Instead, users were only offered hints if they had difficulties. Fortunately, this did not happen, and the testers were able to complete the task without any problems. After the users completed the task and filled in their answers, they were asked for their short feedback on their experience, and notes were taken for further study. During the testing process, the author has also tried to discuss what users would improve or find unintuitive based on their observations and actions.

6.3 Testing results

A better look at the results of user testing visually representing if the test was completed successfully, user ratings of individual processes, and their provided written feedback is provided in Appendix A.

Feature 1 - User registration, profile creation, and login

Success rate: 4/5 testers completed the task.

Average rating: 4.4/5.

Evaluation: The feature generally worked well. Positive feedback appreciated the ease of registration via Google accounts and the synchronization of profile pictures and information. Testers liked the "magic link" login that worked on all devices without requiring a password. However, testers noted areas for improvement: the inability to edit the profile picture directly after creating the profile and the absence of an error message when entering incorrect data.

Key takeaways for improvement:

- Add an option to change the profile picture during the registration step.
- Improve error messaging, e.g., explicitly stating the issue when entering incorrect inputs.

Feature 2 - Create a farm page

Success rate: 4/5 testers completed the task.

Average rating: 4/5.

Evaluation: This feature received mixed responses. Testers appreciated functionalities like automated data filling using links and the integration of Google Maps for location. Nevertheless, several users found it unclear exactly where to start creating a new farm page, suggesting the feature should be more prominently accessible. Testers likewise highlighted the demand for visible responses, like loaders or confirmation messages, which caused uncertainty throughout the process.

The description generation feature was considered promising but underwhelming at times, requiring further refinement for accuracy and relevance.

Key takeaways for improvement:

- Add more apparent navigation options for starting the farm page creation process (e.g., in the sidebar or dropdown menu).
- Improve visual feedback with loaders and confirmation messages.
- Enhance the description generation feature to ensure more consistent quality.
- Redirect users to their new farm page after creation.

Feature 3 - Favorites

Success rate: All users completed the task.

Average rating: 5/5.

Evaluation: This feature was well-received and intuitive. Testers appreciated the clear visibility of liked farms as well as the capability to access favorites via equally the sidebar and profile dropdown menu. The procedure for marking and unmarking favorites worked out great, without functionality issues reported. Some testers suggested adding a notification to confirm when a farm or event is successfully saved to favorites.

Key takeaways for improvement:

- Add a notification to confirm successful additions to the favorites list.

Feature 4 - Review and rating system

Success rate: All users completed the task.

Average rating: 4.2/5.

Evaluation: This feature worked as intended, but feedback indicated some usability issues. Although functionality was operational, some users found the interface could have been clearer and more intuitive.

Key takeaways for improvement:

- Add a "write review" button within the review section.
- Improve visibility and accessibility of the star rating system.
- Introduce Markdown options for review text formatting.
- Implement a character counter or note about the maximum review length.
- Fix delays in displaying submitted reviews.

Feature 5 - Getting in touch with farms

Success rate: All users completed the task.

Average rating: 4.6/5.

Evaluation: This feature was appreciated for its simplicity and functionality. Testers noted the pre-written starter messages, the ability to choose which contact details to provide, and the automated email system. The pop-up notifications confirming message delivery were displayed correctly. Some testers suggested distinguishing required fields more clearly and integrating a complete messaging system for improved interactivity.

Key takeaways for improvement:

- Clearly indicate mandatory and optional fields in the message form.
- Consider implementing an integrated messaging feature within the platform.
- Retain and refine the automated email functionality for ease of use.

Conclusion

The development of the agrotourism platform is a first step towards addressing the digital challenges of the agritourism sector. Based on a careful analysis of user requirements gathered from farm visits and direct interactions with farmers, the application is developed to meet both farmer and visitor needs. The platform is a simple-to-use solution that helps farmers to promote their products and connect with potential visitors.

Although most of the planned functionalities are implemented and operational, some limitations remain and could be improved. Future features, capabilities, and user experience will be refined, and the platform will be expanded. These planned improvements reflect the project team's intent to continue developing the platform, applying experience and knowledge gained during the process to refine techniques and methods.

It must be acknowledged that time and resources slowed the scope of this project and prevented some planned features and advanced functionalities. Further in-depth user feedback and a deeper analysis of broader agritourism trends would help the platform in its future iterations. Nevertheless, these limitations create opportunities for continued development and research.

Finally, the platform shows how technology can provide a functional and impactful solution to connect farmers and visitors. The project addresses immediate challenges in agritourism and prepares the way for future innovation and growth. This thesis results in a resource and room for further exploration and improvement in both technical and user-centric areas so that the platform can grow and change with the user's needs.

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Appendices

Appendix A: User testing results visual representation

Feature 1 - User registration, profile creation, and login

Figure 7: Results of testing Feature 1 (source: author)²

- Visit the application demo web page at <https://farmliaison.vercel.app> - Try to sign in with social media or with email - Complete the sign in steps, n...formation and upload a picture - Save your profile
5 responses

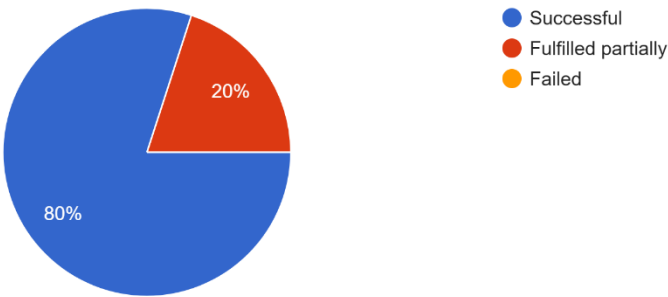
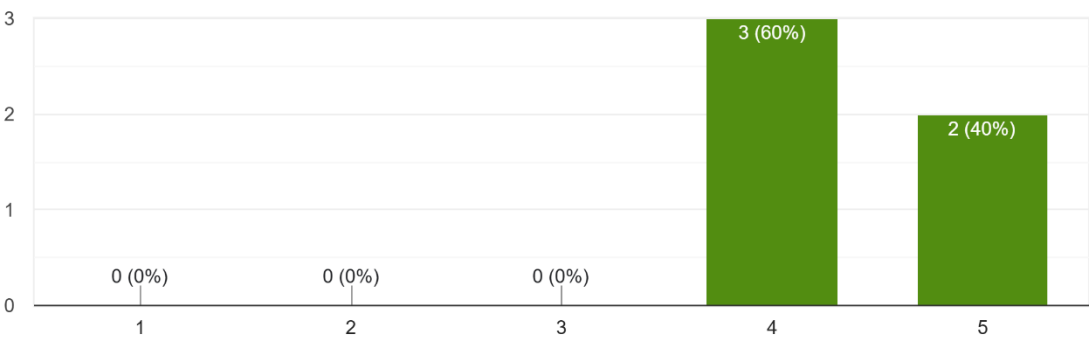


Figure 8: Ratings of testing Feature 1 (source: author)

How was your experience?
5 responses



² During the testing phase, one out of five testers encountered an issue that prevented the completion of the registration process. This issue was traced to an error on the hosting platform, Vercel. The problem was promptly resolved and did not reoccur in subsequent testing sessions.

Written feedback:

- **User 1:** “Error message with text when entering wrong number, not failed to create account”
- **User 2:** “I like how I was able to register quickly with google account and my information and profile picture were automatically uploaded, however I would like to be able to change the picture right in the profile creation menu. I have also tried to login with email from another device and magic link was sent to my email and I was able to login into the same account without the need for password”
- **User 3:** “Registration was quick and smooth. The email confirmation worked perfectly.”
- **User 4:** “I registered with Google, and it was super quick. It even pulled my profile picture and info automatically, which saved time. But I wanted to change my picture right away, and I couldn’t find the option to do that in the profile creation step. Apart from that, everything worked perfectly, including accepting terms and saving my details.”
- **User 5:** “I had registered with Google, and it has worked fine.”

Feature 2 - Create a farm page

Figure 9: Results of testing Feature 2 (source: author)

- Try to create your farm page - As an example you may use <https://fincaterramor.com> link to automatically generate description. google maps link: <https://maps.app.goo.gl/qZdRhUNVuPnzAvNg7>
5 responses

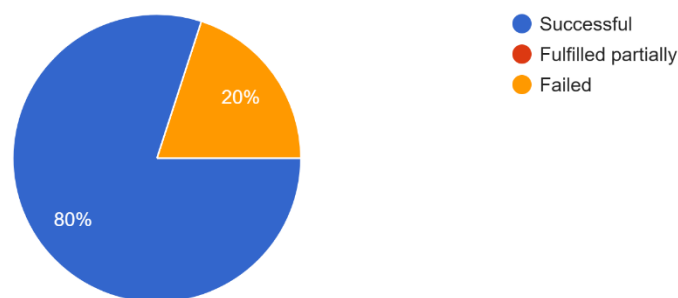
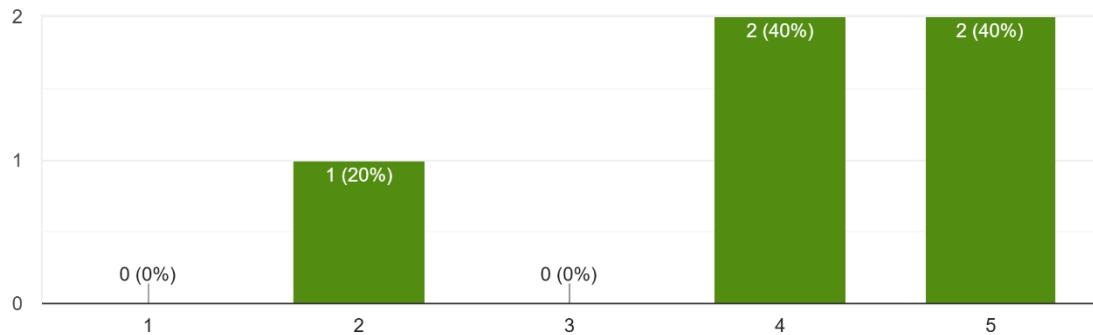


Figure 10: Ratings of testing Feature 2 (source: author)

How was your experience?

5 responses



Written feedback:

- **User 1:** “Not clearly shown what to press to create new farm. I except to have it in profile drop down menu or in side bar. If current user is not Farmer and he tries to create new farm, then show a notification to user, so he changes it in settings.”
- **User 2:** “I really liked how I was able to automatically fill in all the information and generate a description using just the link to farms page (I have found some random farm website on google) the only thing it dint fill automatically was address. Cons - there were no animations or loaders when I pressed any buttons so I could not understand if it worked or not, but after waiting a while everything appeared and my farm page was created.”
- **User 3:** “i really liked the automatic generation of description, however sometimes the results are underwhelming (tested with different farm links), with further tweak-ing it could become a really great feature. UX part was decent, could have further improvement.”
- **User 4:** “I experimented with creating a farm page which seemed like it could be simpler. Add the Google Maps link and it added the location properly (helpful) but the description it generated was weak and ill-advised on what the farm really offers. I was also uncertain if my changes were saved instantly as there was no confirmation or visual feedback after clicking save. It worked overall but could be easier to use.”
- **User 5:** “worked, the description generation is interesting, would be better if i was redirected to my new farm page”

Feature 3 - Favorites and wish list

Figure 11: Results of testing Feature 3 (source: author)

- Go to either Home, Farms or Events tab - Explore farms and events, add a few of them to favourites - Find the Favourites tab and ensure yo...ked farms and events are saved and displayed there
5 responses

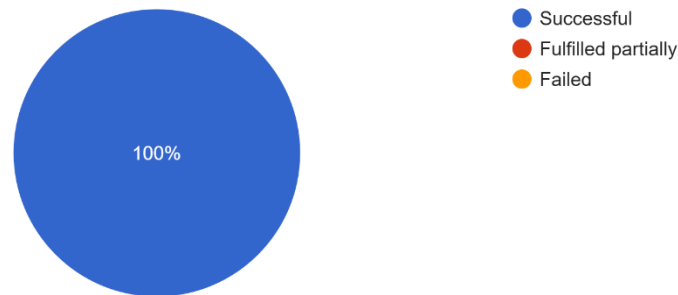
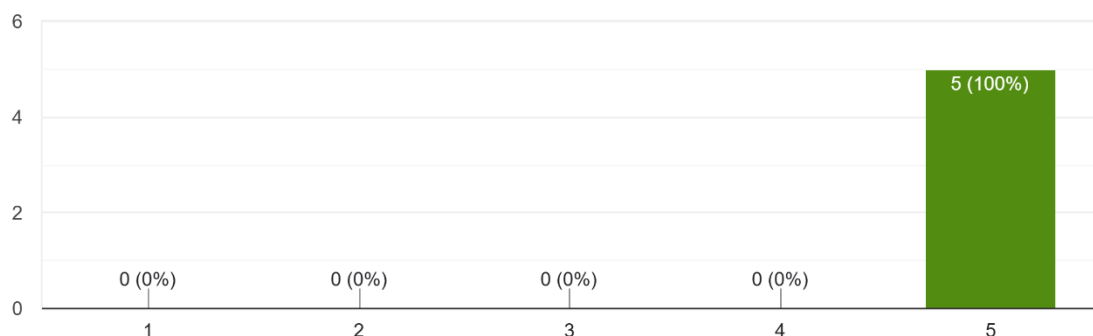


Figure 12: Ratings of testing Feature 3 (source: author)

How was your experience?
5 responses



Written feedback:

- **User 1:** “It is clearly visible what farms are already liked and which ones are not. Good that we have 2 ways to see favorites, by choosing menu in side bar and under profile drop down menu. Personally, I would prefer to have a notification about newly liked farm/event.”
- **User 2:** “Easy to understand how the system works, I am able to mark and un-mark farms as favourites and the favourite tab is always accessible.”
- **User 3:** “Functioning properly, intuitive.”

- **User 4:** “I explored the farms and events and added a few to my favorites. The process was easy, and everything I saved showed up correctly in the Favorites tab.”
- **User 5:** “worked fine, i wish there was a notification to know if it saved successfully”

Feature 4 - Review and rating system

Figure 13: Results of testing Feature 4 (source: author)

- Visit the farm at <https://farmliaison.vercel.app/farm/finca-ecologica-terramor> - Try to leave a review

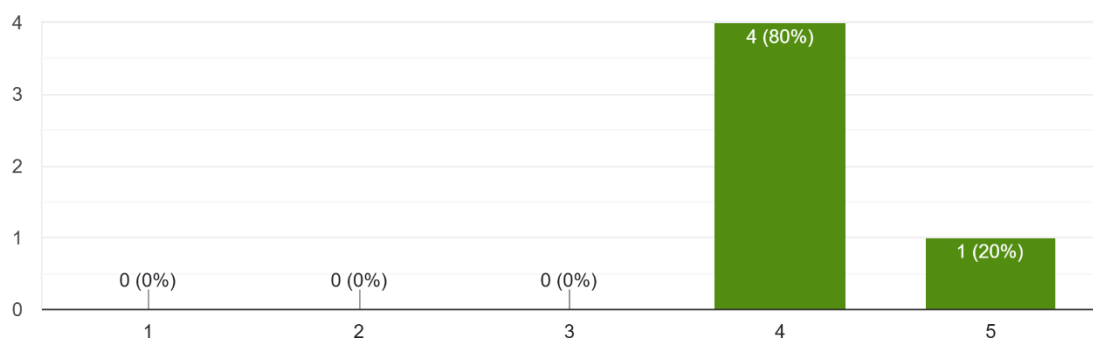
5 responses



Figure 14: Ratings of testing Feature 4 (source: author)

How was your experience?

5 responses



Written feedback:

- **User 1:**
 - “Add review button in review section also
 - Not clearly visible star rating, but pop up notification told me what to do
 - Add Markdown text options”.

- **User 2:** “The overall experience was a bit confusing UI definitely needs improvement for better intuitiveness, however I was able to leave a review.”
- **User 3:** “i had to reload the page for my review to display.”
- **User 4:** “I went to the farm page and left a review without trouble. The process was straightforward but a character counter or a note about the maximum length for reviews would have been good.”
- **User 5:** “took me a moment to figure out how to leave a review but the function is working properly”

Feature 5 - Getting in touch with the farms

Figure 15: Results of testing Feature 5 (source: author)

- Visit the farm at <https://farmliaison.vercel.app/farm/newtown-farm-country-house> - Write a message to the farmer

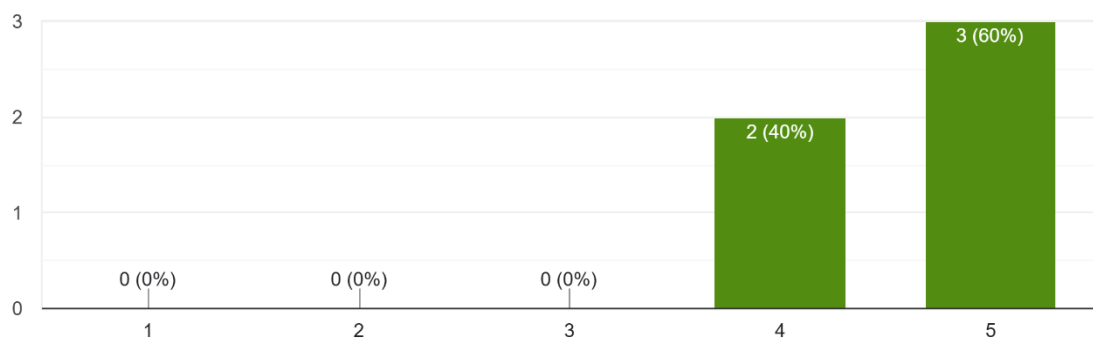
5 responses



Figure 16: Ratings of testing Feature 5 (source: author)

How was your experience?

5 responses



Written feedback:

- **User 1:**
 - “See what field are mandatory and what fields are not.
 - Very nice that example text is already provided for me
 - Pop up notification ensures me that I sent message”.
- **User 2:** “I liked that the starter message was already pre-written and I had a choice of which number I wanted to provide. I also didn’t need to open my email app, create

a message etc. it was all simple and seamless without leaving the app. As Grigorii told me, he successfully received the message to his email (since his farm was used for testing).”

- **User 3:** “Its nice that email is automated, but complete integrated messaging feature would be better.”
- **User 4:** “I visited the farm page and sent a message to the farmer. It was easy to write and send the message.”
- **User 5:** “tested with a test farm, received an automated email with nice styling”

Appendix B: Code and demo application

The application code is available at <https://github.com/gnazarov42/Farmliaison>

The demo of the application is available at <https://farmliaison.vercel.app>